Physics-Informed Learning for Tropical Cyclone Intensity Prediction

Cong Wang, Zhao Chen, *Member, IEEE*, and Fayyaz Minhas

***Abstract*— Accurate prediction of tropical cyclone (TC) intensity is important to disaster prevention. However, many deep learning (DL) models analyzing remote sensing images for TC intensity prediction lack explainability or interpretability due to insufficient consideration of physical knowledge. Therefore, we propose physics-informed cyclone intensity prediction networks (Pici-Nets) to forecast TC intensity measured by the maximum sustained wind (MSW) speed. There are three types of physical information closely related to MSW being used, the structural knowledge of cyclones reflected by multitemporal multispectral images (MSIs), the motional prior representing the dynamics of TC, and the environmental factors affecting the evolvement of TC provided by the public track data. As the multimodal information are fused by Pici-Nets, the MSW speed prediction process which imitates the manual forecast by considering multiple factors to reduce the prediction errors can be regarded as interpretable. Specifically, the first type of information guides Pici-Nets to sharpen the low-spatial-resolution MSIs, enhance temporal structural features, and produce explainable feature maps. Since the second or the third type of data are sometimes missing, we derive Pici-Net+ and Pici-Net++ from the vanilla Pici-Net to cope with the situations. In Pici-Net++, we even incorporate a physics simulation model (PSM) to simulate the missing data. This way, the models become applicable to incomplete datasets and more interpretable in MSW speed prediction. Experiments on different TC datasets validate the efficacy of Pici-Nets, as the results show that Pici-Nets outperform many classic and state-of-the-art (SOTA) models in TC intensity prediction.**

***Index Terms*— Maximum sustained wind (MSW), multimodal fusion, physics simulation model (PSM), tropical cyclone (TC).**

I. Introduction

TROPICAL cyclone (TC) brings natural disasters that can severely damage properties and even result in life causalities [1]. Therefore, it is imperative to predict TC intensity accurately and timely in weather forecasts. The maximum sustained wind (MSW) speed near the center of a cyclone is a critical index for TC intensity. With the development of remote sensing, meteorological/geostationary satellites can now capture temporal multispectral images (MSIs) with high-dimensional information about cyclones, thus enabling accurate MSW speed prediction.

As machine learning (ML), especially deep learning (DL), thrives, it has been largely applied to remote sensing image analysis for TC intensity forecasts. While most of the models for TC MSW prediction are designed primarily around ML and image analysis [2], [3], [4], [5], [6], [7], they lack the physical sense required in meteorological forecasting. They overlook the need to consider multiaspect data including image features, motional priors [8], [9], and environmental factors [10], [11] related to TC evolution. Hence, they are often limited in interpretability or long-term prediction ability. Though there are some methods that do consider various factors of TCs and achieve high prediction accuracy, they require the multimodal information fully provided with the TC images, which, however, cannot always be satisfied in reality [10], [12], [13]. Some of the DL methods are weak in explainability as they do not explicitly represent temporal characteristics of cloud structures such as changes in the eye and the spiral rainbands of TCs over time that are related to MSW speed [14].

To leverage the multimodal data, we propose physics-informed cyclone intensity prediction networks (Pici-Nets). With temporal structural feature enhancement (TSFE) and multimodal physical data fusion (MPDF), Pici-Nets can accurately predict MSW speed in 0–24 h. TSFE exploits structural knowledge of TCs from multiresolution temporal MSIs, highlighting the structural patterns closely related to TC MSW. The MPDF module fuses the temporal structural patterns, the motional prior (e.g., the TC translation speed (km/h) [8], [9]), and the environmental factor, (e.g., the minimum sea level pressure at the TC center (hPa) [10], [11]) for accurate MSW speed prediction. Moreover, we design a physical simulation model (PSM) for Pici-Nets to deal with data dependence and extend the applications of Pici-Nets to incomplete TC datasets. PSM accomplishes the task by simulating the physical parameters lacking by the incomplete TC datasets using the common properties shared by the complete TC datasets. The contributions are as follows.

1) This work proposes a new interpretable network, Pici-Net with its variations Pici-Net+ and Pici-Net++, that imitates human experts making decisions with different perspectives and facts. They incorporate the multimodal physical information into deep networks to enhance TC MSIs and reduce MSW speed errors in 24-h prediction.
2) This work also looks into the explainability of the DL models for MSW speed prediction. The proposed Pici-Nets provide users with rich intermediate results

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such as cyclone eye features and cloud spiral textures that are physically meaningful and human-friendly to better facilitate meteorological analysis for TC forecast.

3) Though leveraging multimodal data, Pici-Nets are not limited by data availability but have wide applications, since Pici-Net+ and Pici-Net++ are generalized to incomplete TC datasets. Skipping the multiresolution MSI enhancement or simulating the motional priors or the environmental factors by PSM, the upgraded models can still accurately predict MSW speed.

For model evaluation, we use the public datasets, TCIR [15] and HURSAT-B1 [16] as well as a self-collected big dataset FY4A-TC with MSIs captured by the FY4A satellite [17] from 2018 to 2023. The FY4A dataset is the most complete with all types of information needed by the vanilla Pici-Net. The other two are somewhat incomplete, as TCIR lacks the multiresolution MSIs while HURSAT-B1 lacks the multiresolution MSIs, the TC translation speed, and the minimum sea level pressure. Experimental results show that Pici-Nets give state-of-the-art (SOTA) performances in 0–24-h MSW speed prediction on variable datasets.

The rest of this article is organized as follows. Section II introduces the related works. Section III presents the proposed Pici-Nets. Section V analyzes and discusses the experimental results. Finally, Section VI concludes the article.

II. Related Works

As TCs have huge impacts on human life, there is a long history of TC forecasts. The approaches include traditional cloud image analysis [18], [19], [20], [21], [22], [23], [24], [25], [26], dynamical forecasting analysis [27], [28], [29], statistical estimation, [30], [31], [32], [33], [34], [35], and ML [2], [3], [4], [5], [6], [7], [33], [36], [37]. With the boom of artificial intelligence (AI), deep neural networks (DNNs), especially convolutional neural networks (CNNs), have caught large attention in the society of TC intensity forecast. While TCs are complex systems, DNNs exhibit remarkable flexibility and nonlinear representation power to extract complex features from the TC data, enabling accurate TC intensity prediction from a view of data mining and image analysis, even without any input of meteorological or other physical knowledge. Thus, this work focuses on the area of DL for TC forecast and proposes a new intelligent method for accurate TC MSW speed prediction. The major approaches in the area are summarized in three categories and introduced as follows.

This first category mainly uses numerical meteorological data. Jin et al. [10] applied DNNs as nonlinear predictors to various meteorological data for tracking and forecasting TC in the short term. Huang et al. [38] utilized the latitude and longitude of cyclone centers, the minimum pressure, and the MSW speed to construct a spatial-temporal network for TC prediction. While being relatively easy to implement, these models could not be conveniently generalized to new forms of TC data such as remote sensing images.

As remote sensing develops, visible (VIS) images, infrared (IR) images, and MSIs captured by meteorological satellites start to show great value in TC analysis. Thus, the community of TC intensity estimation (TCIE) and prediction start to show interest in spectral image analysis. Chen et al. [2], [3] utilized satellite infrared brightness temperature and microwave rain-rate data, employing CNN to predict TC intensity, besides postsmoothing processing is applied to further improve the prediction accuracy. Similarly, Dawood et al. [4] proposed a deep CNN, Deep-PHURIE, for TC intensity prediction, which is robust to shifts or errors in the center of annotated TC in the images after extensive benchmarking experiments. Building on these efforts, Zhang et al. [5] proposed a cascaded CNN model that first classified TCs by their intensity levels through a TC intensity classification (TCIC) module and then predicted MSW speed based on the classification results through a TCIE module. However, as these models were designed for VIS or IR images, they could only extract spatial features of satellite images to predict intensity, not being able to utilize the spectral features in MSIs, which are more advanced and informative than the VIS or IR images. To address the issue, Chen and Yu [6] proposed a tensor-based CNN (T-TCNN) consisting of a classifier for intensity categorization and a regressor for wind speed estimation. While it can extract spectral–spatial information for TC image analysis from MSIs by tensor decomposition, it adopts a multistage strategy that may lead to error accumulation. To capture comprehensive features, Xiu et al. [7] utilized a spectral–temporal dislocation (STD-AN) technique and vision transformer to learn the temporal–spatial–spectral features of TC to predict TC intensity. Wang et al. [37] fused the multiresolution MSIs to generate high-spatial-resolution images and predicted TC intensity with the sharpened images by a deep network incorporated with Tucker decomposition, a tensor analysis model. However, these methods do not take into account the complex motional priors describing the dynamics of TC or the environmental factors affecting the evolvement of TC during the prediction process, thus lacking interpretability.

With the advent of meteorological big data, DL methods that exploit both environmental factors and TC MSIs for cyclone intensity prediction have emerged. Boussioux et al. [12] fed latitude, longitude, minimum pressure at the center of TC, distance-to-land, and other environmental factors along with TC MSIs to an encoder–decoder and predicted TC intensity. Similarly, Li et al. [13] proposed a multimodal adversarial fusion (MAF) model, which used the TC latitude, longitude, the minimum pressure at the center of TC, the wind speed, and the wind field data at multiple timestamps as observational data. The MAF model employed an attention mechanism to achieve multimodal fusion of these observational data with the TC MSIs, achieving excellent intensity prediction results. However, these models could only perform on the datasets provided with all the required data while overlooking the temporal characteristics of cloud structures closely related to MSW. Any missing data could paralyze the models. Therefore, not only do we need to utilize the multimodal data of TCs, but also reducing the dependency on data forms to improve the generalizability of the models for smart TC MSW speed prediction.

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III. Methods

To improve TC MSW prediction, especially tracking TC for longer hours, we exploit multimodal data and physical knowledge of TC, thereby proposing Pici-Net (the vanilla model). To accommodate a variety of incomplete datasets, we derive Pici-Net+ and Pici-Net++ from Pici-Net. These three models are collectively referred to as Pici-Nets. The vanilla Pici-Net mainly performs TSFE and MPDF before MSW prediction, as illustrated in Fig. 1. TSFE consists of a number of parts, a multiresolution fusion module that merges low-resolution MSIs (LR-MSIs) and high-resolution MSIs (HR-MSIs) to generate high-quality MSIs, an eye-spiral feature extractor that learns temporal structural patterns of TC, a structural information injector that integrates the eye-spiral features into spectral–spatial features, and a temporal-structural feature extractor that learns the structural difference between two sets of high-quality MSIs, one reconstructed from the present MSIs and the other from the past. Guided by the structural knowledge, TSFE enhances the quality of TC MSIs and obtains physically-meaningful image features for MSW prediction. Meanwhile, MPDF combines the temporal image features, the TC structural textures, the motional prior, and the environmental factor to depict the dynamic patterns of TCs. As the multiresolution MSIs may not be provided in some datasets (i.e., TCIR [15] and HURSAT-B1 [16]), we modify TSFE in Pici-Net and obtain Pici-Net+ as illustrated in Fig. 2. For the datasets without the motional prior or the environmental factor, we design PSM to simulate these physical data as illustrated in Fig. 3 and build Pici-Net++ to predict MSW speed as illustrated in Fig. 4. The details of the proposed networks are described as follows.

A. Pici-Net

The inputs and the outputs for Pici-Net are defined as follows. Let $\underline{\mathbf{X}}_{i,t} \in \mathbb{R}^{h\times w\times B}$ and $\underline{\mathbf{X}}_{i,t-n} \in \mathbb{R}^{h\times w\times B}$ denote LR-MSIs of the ith TC sampled at the tth and the $t-n$th timestamps, respectively, where $t = n, \ldots, T$, n is the time lag empirically defined for Pici-Nets to exploit historical data, T is the max existing time of the ith TC, and $i = 1, \ldots, I$, I is the number of TCs. Likewise, $\underline{\mathbf{Y}}_{i,t} \in \mathbb{R}^{H\times W\times b}$ and $\underline{\mathbf{Y}}_{i,t-n} \in \mathbb{R}^{H\times W\times b}$ denote HR-MSIs of a TC at the tth time and the $t-n$th time, respectively, where $b < B, h < H$ and $w < W$. The motional prior is the TC translation speed $\mathrm{s}(\underline{\mathbf{X}}_{i,t})$, calculated based on the latitude and the longitude coordinates of the TC MSIs $\underline{\mathbf{X}}_{i,t}$ which are provided in the track data released by weather forecast administrations. The environmental factor is the sea level pressure at the TC center $p(\underline{\mathbf{X}}_{i,t})$ of $\underline{\mathbf{X}}_{i,t}$ also provided by the track data. The final output is the predicted MSW speed value $\hat{\mathrm{o}}_{t+m} \in \mathbb{R}$ for the MSIs $\underline{\mathbf{X}}_{i,t+m}$ of the ith TC at some future timestamp $t+m$, where $m = 0, 6, 12, 18, 24$ h representing the lapse of time. The reference MSW speed value $\mathrm{o}_{i,t+m} \in \mathbb{R}$ used as the training label is obtained from the track data. As can be seen from Fig. 1, Pici-Net performs TSFE, MPDF, and MSW speed prediction, realized by modules (a)–(d), module (e), and module (f), respectively. The modules are described as follows.

1) Multiresolution Fusion Module: The multiresolution fusion module extracts features from LR-MSIs and HR-MSIs through two parallel branches and then fuses these features to obtain high-quality MSIs with both high spatial and high spectral resolutions. As shown in Fig. 1(a), initially, we upsample $\underline{\mathbf{X}}_{i,t} \in \mathbb{R}^{h\times w\times B}$ to $\underline{\mathbf{X}}_{i,t}^{(\mathrm{Upsampled})} \in \mathbb{R}^{H\times W\times B}$ via bilinear interpolation to match the spatial size of $\underline{\mathbf{Y}}_{i,t}$. Then, we encode the features of the original MSIs $\underline{\mathbf{X}}_{i,t}^{(\mathrm{Upsampled})}$ and $\underline{\mathbf{Y}}_{i,t}$ by feeding them to three feature extraction blocks (colored in blue in Fig. 1) in the parallel branches. Each feature extraction block consists of 2-D convolution, batch-normalization, and activation. For dimensionality reduction and efficiency, we attach max-pooling to some of the feature extraction blocks as illustrated in Fig. 1(a). The directly encoded features are denoted by $\underline{\mathbf{X}}_{i,t}^{(1)}, \underline{\mathbf{X}}_{i,t}^{(2)}$ while the cross-encoded features $\mathbf{Y}_{i,t}^{(1)}$ and $\underline{\mathbf{Y}}_{i,t}^{(2)}$ are obtained by feeding the intermediate representations in one branch to the feature extractor in the other branch. During the decoding stage that reconstructs the high-quality MSIs, we feed the encoded features to a multiscale feature fusion block and obtain

$$\underline{\mathbf{Z}}_{i,t}^{(1)} = \mathbf{F}^{(\mathrm{Concat})}\left(\left(\left(\underline{\mathbf{X}}_{i,t}^{(1)} + \underline{\mathbf{Y}}_{i,t}^{(2)}\right), \left(\underline{\mathbf{X}}_{i,t}^{(2)} + \underline{\mathbf{Y}}_{i,t}^{(1)}\right)\right)\right) \quad (1)$$

where $\mathbf{F}^{(\mathrm{Concat})}(\bullet)$ is the concatenation function. We continue to upsample $\underline{\mathbf{Z}}_{i,t}^{(1)}$ and reconstruct MSIs $\underline{\mathbf{Z}}_{i,t}^{(\mathrm{Fuse})} \in \mathbb{R}^{H\times W\times B}$ via three layers 2-D transposed convolution. It can be seen that $\underline{\mathbf{Z}}_{i,t}^{(\mathrm{Fuse})}$ combines the merits of both inputs, $\underline{\mathbf{X}}_{i,t}$ and $\underline{\mathbf{Y}}_{i,t}$, adopting the highest spatial resolution and the highest spectral resolution of the original MSIs.

2) Eye-Spiral Pattern Descriptor: Considering the close relationship between TC structure and intensity [14], we leverage the structural knowledge and extract features from the eye region and the cloud spiral of each TC sample to boost MSW prediction accuracy. Following the method used in [39], we define the central area of a TC image as the eye region. This practical method is also feasible for TC MSIs not showing clear eyes, as the central features of TCs are related to TC intensity [19]. Specifically, we select the last band $\mathbf{Y}_{i,t}^{(\mathrm{Last})} \in \mathbb{R}^{H\times W}$ of HR-MSIs $\underline{\mathbf{Y}}_{i,t}$ to acquire the eye regions and recognize the eye-spiral patterns for two reasons. First, HR-MSIs contain richer spatial information than LR-MSIs. Second, as sampled via the infrared channel, the last band of MSIs is usually of good quality, not affected by sampling time (during the night when there is no sunlight, the VIS channels cannot work properly, resulting in noisy or invalid band data). In each $\mathbf{Y}_{i,t}^{(\mathrm{Last})}$, we draw the central $h \times w$ part as the region of interest (ROI) for the TC eye, denoted as $\mathbf{Y}_{i,t}^{(\mathrm{Eye})} \in \mathbb{R}^{h\times w}$. We designate the remaining part of $\mathbf{Y}_{i,t}^{(\mathrm{Last})}$ as the spiral ROI, denoted by $\mathbf{Y}_{i,t}^{(\mathrm{Spiral})} \in \mathbb{R}^{H\times W}$, with the eye ROI padded by 0, as shown in Fig. 1(b).

To extract spiral structural features, we feed $\mathbf{Y}_{i,t}^{(\mathrm{Spiral})}$ to two parallel feature extraction blocks (colored in blue), obtain query $\mathbf{Q}_{i,t}^{(\mathrm{Spiral})}$ and key $\mathbf{K}_{i,t}^{(\mathrm{Spiral})}$ reshaped to size $HW \times B$ and input them to a self-attention block, which, adopted from reference [40], computes the spiral feature map as follows:

$$\mathbf{Y}_{i,t}^{(\mathrm{Spiral\text{-}feature})} = \mathbf{F}^{(\mathrm{Softmax})}\left(\mathbf{Q}_{i,t}^{(\mathrm{Spiral})} \cdot \mathbf{K}_{i,t}^{(\mathrm{Spiral})^{\mathrm{T}}} / \sqrt{d_1}\right) \quad (2)$$

where $\mathbf{Y}_{i,t}^{(\mathrm{Spiral\text{-}feature})} \in \mathbb{R}^{HW\times HW}$, $d_1 = HW$, $\mathbf{F}^{(\mathrm{Softmax})}(\bullet)$ is the Softmax function for matrices and T represents the transpose.

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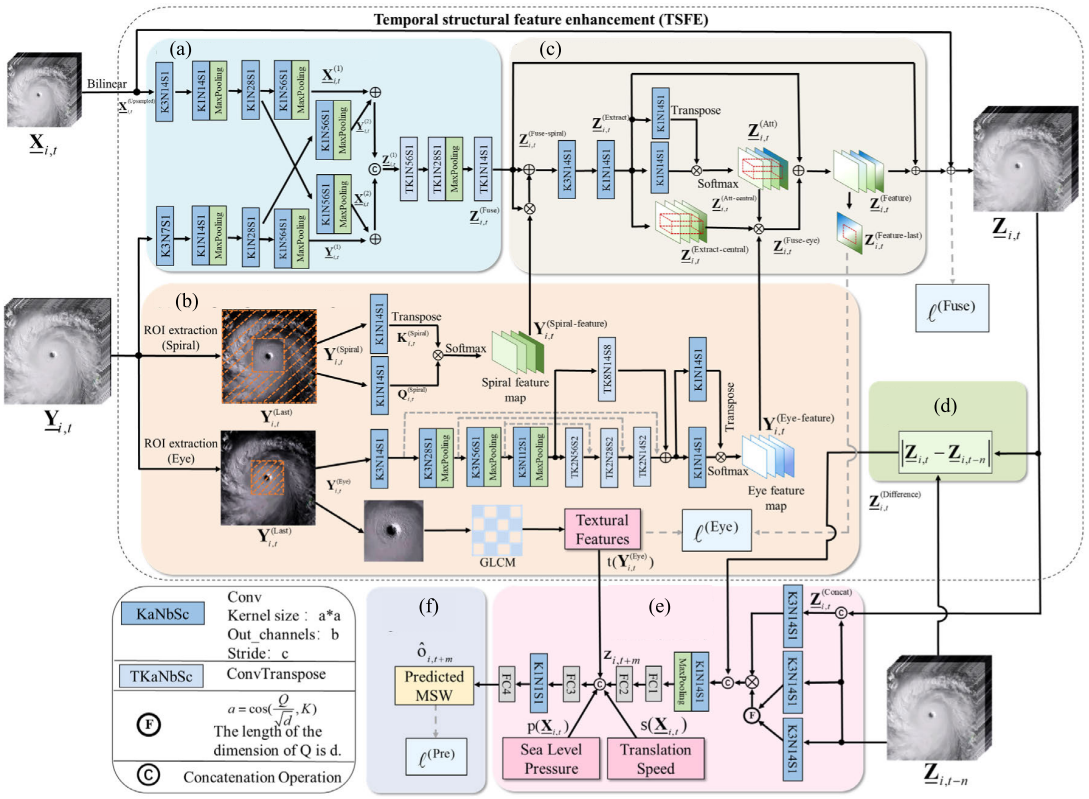


Fig. 1. Proposed Pici-Net for TC MSW prediction. (a) Multiresolution fusion module. (b) Eye-spiral pattern descriptor. (c) Structural information injector. (d) Temporal-structural feature extractor. (e) MPDF. (f) MSW prediction. TSFE is composed of modules (a)–(d). The blue, green, and gray blocks represent convolutional layers, max-pooling layers, and fully connected layers, respectively.

For the eye region, we employ both deep feature representation and hand-crafted feature extraction. First, we input $\mathbf{Y}_{i,t}^{(\text{Eye})}$ to an encoder–decoder architecture comprising a U-Net [41] and a local skip connection, and result in $\underline{\mathbf{Y}}_{i,t}^{(\text{Eye-coder})} \in \mathbb{R}^{h\times w\times B}$. Then, we feed $\underline{\mathbf{Y}}_{i,t}^{(\text{Eye-coder})}$ to two parallel feature extraction blocks, obtain query $\mathbf{Q}_{i,t}^{(\text{Eye})}$ and key $\mathbf{K}_{i,t}^{(\text{Eye})}$ reshaped to size $hw \times B$, and input them to the self-attention block [40] of the same architecture as defined in (2)

$$\mathbf{Y}_{i,t}^{(\text{Eye-feature})} = \mathbf{F}^{(\text{Softmax})}\left(\mathbf{Q}_{i,t}^{(\text{Eye})} \cdot K_{i,t}^{(\text{Eye})^{\text{T}}} / \sqrt{d_2}\right) \tag{3}$$

where $\mathbf{Y}_{i,t}^{(\text{Eye-feature})} \in \mathbb{R}^{hw\times hw}$ and $d_2 = hw$. Since we need to predict MSW speed near the center of TC, the information from the Eye ROI is crucial. To improve the explainability of Pici-Net, we extract hand-crafted textural features from the gray-level co-occurrence matrix (GLCM) [42], [43], [44] of the Eye ROI, in complement to the Eye ROI features learned by the network. Specifically, we sum up entropy, homogeneity, energy, contrast, and correlation of GLCM of $\mathbf{Y}_{i,t}^{(\text{Eye})} \in \mathbb{R}^{h\times w}$ to obtain the textural feature $\text{t}\left(\mathbf{Y}_{i,t}^{(\text{Eye})}\right)$ [39], [44].

3) Structural Information Injector: To enhance structural information and fusion quality, we design an injector adapted from the attention mechanism [40] as illustrated in Fig. 1(c). It injects eye region and spiral information encoded in Fig. 2(b) to the fused high-quality MSIs produced in Fig. 1(a). To reuse features and prevent information loss in representations, we attach local and global skip connections to the injector. The former enables a short-distance information transfer while the latter enables a long-range one. The local skip connections bring the output of one layer directly to a subsequent layer, improving the efficiency of feature transfer within the network. Global skip connections span multiple layers in the network, connecting the output of the input layer or shallow layers directly to a deep layer in the network, thereby preserving details from the features learned many layers before or even the original input data. With the skip connections, the TC MSIs are further enhanced with the eye-spiral details injected.

First, we reshape $\underline{\mathbf{Z}}_{i,t}^{(\text{Fuse})} \in \mathbb{R}^{H\times W\times B}$ output by (1) to $\mathbf{Z}_{i,t}^{(\text{Fuse})} \in \mathbb{R}^{HW\times B}$ and combine it with the spiral features $\mathbf{Y}_{i,t}^{(\text{Spiral-feature})} \in \mathbb{R}^{HW\times HW}$ output by (2) to obtain the MSIs $\mathbf{Z}_{i,t}^{(\text{Fuse-spiral})} \in \mathbb{R}^{HW\times B}$ with enhanced spiral structures as follows:

$$\mathbf{Z}_{i,t}^{(\text{Fuse-Spiral})} = \mathbf{Y}_{i,t}^{(\text{Spiral-feature})} \cdot \mathbf{Z}_{i,t}^{(\text{Fuse})}. \tag{4}$$

Next, we inject eye information and enhance eye features. We feed $\underline{\mathbf{Z}}_{i,t}^{(\text{Fuse-spiral})} \in \mathbb{R}^{H\times W\times B}$ reshaped from $\mathbf{Z}_{i,t}^{(\text{Fuse-spiral})}$ to two feature extraction blocks and obtain the representation $\underline{\mathbf{Z}}_{i,t}^{(\text{Extract})} \in \mathbb{R}^{H\times W\times B}$, which is then input to the self-attention as in (3) and [40] and end up as attention maps $\underline{\mathbf{Z}}_{i,t}^{(\text{Att})} \in \mathbb{R}^{HW\times HW}$. We notice that $\underline{\mathbf{Z}}_{i,t}^{(\text{Extract})}$ and $\underline{\mathbf{Z}}_{i,t}^{(\text{Att})}$ also contain eye features. As the eye region is crucial to TC intensity prediction, it is better to leverage all types of eye features from different perspectives. Therefore, we draw the central part $\underline{\mathbf{Z}}_{i,t}^{(\text{Extract-central})} \in \mathbb{R}^{h\times w\times B}$ from $\underline{\mathbf{Z}}_{i,t}^{(\text{Extract})}$ and $\underline{\mathbf{Z}}_{i,t}^{(\text{Att-central})} \in \mathbb{R}^{hw\times hw}$ from $\underline{\mathbf{Z}}_{i,t}^{(\text{Att})}$ and combine them with $\mathbf{Y}_{i,t}^{(\text{Eye-feature})} \in \mathbb{R}^{hw\times hw}$ output by (3) to obtain the integrated eye features $\underline{\mathbf{Z}}_{i,t}^{(\text{Eye})} \in \mathbb{R}^{h\times w\times B}$ as follows:

$$\underline{\mathbf{Z}}_{i,t}^{(\text{Eye})} = \text{reshape}\left(\mathbf{Y}_{i,t}^{(\text{Eye-feature})} \mathbf{Z}_{i,t}^{(\text{Att-central})T} \mathbf{Z}_{i,t}^{(\text{Extract-central})}\right) \tag{5}$$

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where $\mathbf{Z}_{i,t}^{(\text{Extract-central})} \in \mathbb{R}^{hw\times B}$ is reshaped from $\underline{\mathbf{Z}}_{i,t}^{(\text{Extract-central})}$. After that, we combine $\underline{\mathbf{Z}}_{i,t}^{(\text{Eye})}$ and the spiral portion of $\underline{\mathbf{Z}}_{i,t}^{(\text{Extract})}$, resulting in $\underline{\mathbf{Z}}_{i,t}^{(\text{Fuse-eye})} \in \mathbb{R}^{H\times W\times B}$. Subsequently, we use a local skip connection to inject $\underline{\mathbf{Z}}_{i,t}^{(\text{Fuse-eye})}$ to $\underline{\mathbf{Z}}_{i,t}^{(\text{Extract})}$ which already carries the spiral structural information as follows:

$$\underline{\mathbf{Z}}_{i,t}^{(\text{Feature})} = \underline{\mathbf{Z}}_{i,t}^{(\text{Fuse-eye})} + \underline{\mathbf{Z}}_{i,t}^{(\text{Extract})}. \tag{6}$$

To align the different representations of the TC eye region and minimize the discrepancy in structural information, we design an eye regularization that connects the descriptor and the injector. We extract the Eye ROI $\mathbf{Z}_{i,t}^{(\text{Feature-eye})} \in \mathbb{R}^{h\times w}$ from the last band of $\underline{\mathbf{Z}}_{i,t}^{(\text{Feature})}$ [39], compute the textural features $\text{t}(\mathbf{Z}_{i,t}^{(\text{Feature-last})})$, compare it with the eye textural features $\text{t}(\mathbf{Y}_{i,t}^{(\text{Eye})})$ obtained by the descriptor and formulate the eye regularization as

$$\ell^{(\text{Eye})} = \frac{1}{I(T-m-n)}\sum_{i=1}^{I}\sum_{t=n}^{T-m}\left|\text{t}\left(\mathbf{Z}_{i,t}^{(\text{Feature-last})}\right) - \text{t}\left(\mathbf{Y}_{i,t}^{(\text{Eye})}\right)\right|. \tag{7}$$

Finally, we use two global skip connections to further enhance the high spatial and high spectral MSIs after the eye-spiral details are injected. As can be seen in Fig. 1(c), the inner-module global skip connection reminds the network of the initially fused MSIs $\underline{\mathbf{Z}}_{i,t}^{(\text{Fuse})}$ and the cross-module global skip connection brings back the original LR-MSIs upsampled to $\underline{\mathbf{Z}}_{i,t}^{(\text{Upsampled})}$. Thus, the output of the injector is

$$\underline{\mathbf{Z}}_{i,t} = \underline{\mathbf{Z}}_{i,t}^{(\text{Feature})} + \underline{\mathbf{Z}}_{i,t}^{(\text{Fuse})} + \underline{\mathbf{X}}_{i,t}^{(\text{Upsampled})}. \tag{8}$$

It can be seen that $\underline{\mathbf{Z}}_{i,t}$ maintains the spectral–spatial features and the enhanced structural details of the TC MSIs.

To train the injector and merge the multiple features of TCs, we introduce a fusion loss as follows:

$$\ell^{(\text{Fuse})} = 0.5\ell^{(\text{Spat}_1)} + 0.5\ell^{(\text{Spat}_2)} + \ell^{(\text{Spec})} \tag{9}$$

where $\ell^{(\text{Spat}_1)}$ and $\ell^{(\text{Spat}_2)}$ are the spatial edge loss, while $\ell^{(\text{Spec})}$ is the spectral edge loss [37], [45], computed as follows:

$$\ell^{(\text{Spat}_1)} = \begin{cases} \displaystyle\sum_{i=1}^{I}\sum_{t=n}^{T-m}\frac{\left\|\mathbf{F}^{(\text{Spat}_1)}\left(\underline{\mathbf{Z}}_{i,t}\right) - \mathbf{F}^{(\text{Spat}_1)}\left(\underline{\mathbf{Y}}_{i,t}\right)\right\|_{\mathbf{F}}^2}{I(T-m-n)bW(H-1)}, & C\in\Omega_1 \\ \displaystyle\sum_{i=1}^{I}\sum_{t=n}^{T-m}\frac{\left\|\mathbf{F}^{(\text{Spat}_1)}\left(\underline{\mathbf{Z}}_{i,t}\right) - \mathbf{F}^{(\text{Spat}_1)}\left(\underline{\mathbf{X}}_{i,t}^{(\text{upsampled})}\right)\right\|_{\mathbf{F}}^2}{I(T-m-n)(B-b)W(H-1)}, & C\in\Omega_2 \end{cases} \tag{10}$$

$$\ell^{(\text{Spat}_2)} = \begin{cases} \displaystyle\sum_{i=1}^{I}\sum_{t=n}^{T-m}\frac{\left\|\mathbf{F}^{(\text{Spat}_2)}\left(\underline{\mathbf{Z}}_{i,t}\right) - \mathbf{F}^{(\text{Spat}_2)}\left(\underline{\mathbf{Y}}_{i,t}\right)\right\|_{\mathbf{F}}^2}{I(T-m-n)bH(W-1)}, & C\in\Omega_1 \\ \displaystyle\sum_{i=1}^{I}\sum_{t=n}^{T-m}\frac{\left\|\mathbf{F}^{(\text{Spat}_2)}\left(\underline{\mathbf{Z}}_{i,t}\right) - \mathbf{F}^{(\text{Spat}_2)}\left(\underline{\mathbf{X}}_{i,t}^{(\text{upsampled})}\right)\right\|_{\mathbf{F}}^2}{I(T-m-n)(B-b)H(W-1)}, & C\in\Omega_2 \end{cases} \tag{11}$$

$$\ell^{(\text{Spec})} = \begin{cases} \displaystyle\sum_{i=1}^{I}\sum_{t=n}^{T-m}\frac{\left\|\mathbf{F}^{(\text{Spec})}\left(\underline{\mathbf{Z}}_{i,t}\right) - \mathbf{F}^{(\text{Spec})}\left(\underline{\mathbf{Y}}_{i,t}\right)\right\|_{\mathbf{F}}^2}{I(T-m-n)WH(b-1)}, & C\in\Omega_1 \\ \displaystyle\sum_{i=1}^{I}\sum_{t=n}^{T-m}\frac{\left\|\mathbf{F}^{(\text{Spec})}\left(\underline{\mathbf{Z}}_{i,t}\right) - \mathbf{F}^{(\text{Spec})}\left(\underline{\mathbf{X}}_{i,t}^{(\text{upsampled})}\right)\right\|_{\mathbf{F}}^2}{I(T-m-n)WH(B-b-1)}, & C\in\Omega_2 \end{cases} \tag{12}$$

where $\|\bullet\|_F$ is the Frobenius norm, C is the band index, $\Omega_1 = \{1,\ldots,b\}$ and $\Omega_2 = \{b+1,\ldots,B\}$. Functions $\mathbf{F}^{(\text{Spat}_1)}(\bullet)$, $\mathbf{F}^{(\text{Spat}_2)}(\bullet)$, and $\mathbf{F}^{(\text{Spec})}(\bullet)$ perform edge mapping along the height, the width and the spectrum, respectively, defined as follows:

$$\mathbf{F}_{h,w,b}^{(\text{Spat}_1)}(\underline{\mathbf{A}}) = \underline{\mathbf{A}}_{h+1,w,b} - \underline{\mathbf{A}}_{h,w,b} \tag{13}$$

$$\mathbf{F}_{h,w,b}^{(\text{Spat}_2)}(\underline{\mathbf{A}}) = \underline{\mathbf{A}}_{h,w+1,b} - \underline{\mathbf{A}}_{h,w,b} \tag{14}$$

$$\mathbf{F}_{h,w,b}^{(\text{Spec})}(\underline{\mathbf{A}}) = \underline{\mathbf{A}}_{h,w,b+1} - \underline{\mathbf{A}}_{h,w,b} \tag{15}$$

where $\underline{\mathbf{A}}$ stands for the unput argument. By the edge losses, Pici-Net can reconstruct MSIs with eye-spiral structural details that inherit the features from the original MSIs but exhibit better quality than the original LR-MSIs or HR-MSIs.

4) Temporal-Structural Feature Extractor: We not only fuse the multiresolution MSIs from the current time t but also incorporate the MSIs from the time stamp $t-n$ to learn temporal-structural features and boost MSW speed prediction accuracy. Similar to the computation of $\underline{\mathbf{Z}}_{i,t}$, we input $\underline{\mathbf{X}}_{i,t-n}$ and $\underline{\mathbf{Y}}_{i,t-n}$ to modules (a)–(c) of Pici-Net to obtain $\underline{\mathbf{Z}}_{i,t-n}$ as illustrated in Fig. 1. Subsequently, we compute the difference between $\underline{\mathbf{Z}}_{i,t}$ and $\underline{\mathbf{Z}}_{i,t-n}$ as the temporal-structural features

$$\underline{\mathbf{Z}}_{i,t}^{(\text{Difference})} = \left|\underline{\mathbf{Z}}_{i,t} - \underline{\mathbf{Z}}_{i,t-n}\right|. \tag{16}$$

5) Multimodal Physical Data Fusion: As the motional priors represent the dynamics of TCs and the environmental factors affect the evolvement of TCs, we integrate the multimodal physical data into the structurally enhanced high-quality MSIs for accurate MSW speed prediction, as illustrated by Fig. 1(e). Given the TC translation speed $\text{s}(\underline{\mathbf{X}}_{i,t})$ positively correlated with TC intensity [8] and the sea level pressure $\text{p}(\underline{\mathbf{X}}_{i,t})$ at the TC center negatively correlated [46] with TC intensity, we combine them with the textural features $\text{t}(\underline{\mathbf{X}}_{i,t})$ extracted from the TC eye region to guide MSW speed prediction.

As the start of the multimodal fusion module, to guarantee high prediction accuracy, we introduce temporal image features by concatenating $\underline{\mathbf{Z}}_{i,t}$ and $\underline{\mathbf{Z}}_{i,t-n}$ which results in $\underline{\mathbf{Z}}_{i,t}^{(\text{Concat})}$. To further exploit the temporal features, we design a temporal self-attention block embedded with cosine similarity. Unlike the original attention mechanism [40] which adopts Softmax for feature map computation, what we use here employs cosine similarity to upgrade efficiency. As illustrated by Fig. 1(e), the modified attention is calculated as

$$\underline{\mathbf{Z}}_{i,t}^{(\text{Feature-time})} = \mathbf{F}^{(\text{Cos})}\left(\underline{\mathbf{Q}}_{i,t-n}/\sqrt{d_3}, \underline{\mathbf{K}}_{i,t-n}\right)\odot\underline{\mathbf{V}}_{i,t} \tag{17}$$

where query $\underline{\mathbf{Q}}_{i,t-n}$ and key $\underline{\mathbf{K}}_{i,t-n}$ are obtained from $\underline{\mathbf{Z}}_{i,t-n}$, value $\underline{\mathbf{V}}_{i,t}$ is obtained from $\underline{\mathbf{Z}}_{i,t}^{(\text{Concat})}$, $d_3 = \text{HWB}$, $\mathbf{F}^{(\text{Cos})}(\bullet)$ is the cosine similarity, and $\odot$ is the Hadamard product. After

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Algorithm 1 Optimization of Pici-Net

Input: the TC MSIs, the TC translation speed, the minimum sea level pressure at the TC center, and the MSW speed $\left\{\underline{\mathbf{X}}_{i,t}, \underline{\mathbf{X}}_{i,t-n}, \underline{\mathbf{Y}}_{i,t}, \underline{\mathbf{Y}}_{i,t-n}, \mathrm{s}\left(\underline{\mathbf{X}}_{i,t}\right), \mathrm{p}\left(\underline{\mathbf{X}}_{i,t}\right), \mathrm{o}_{i,t+m}\right\}_{i=1,t=n}^{I,T-m}$;
Output: the model $\Psi_1(\cdot, \Theta_1)$ and the predicted MSW speed $\left\{\hat{\mathrm{o}}_{i,t+m}\right\}_{i=1,t=n}^{I,T-m}$;
Steps:

1. Initialize Θ_1;
2. **While** Θ_1 is not converged **do**
3. **for** $i = 1, 2, \ldots, I, t = n, n+1, \ldots, T-m$
   // TSFE
4. Feed $\underline{\mathbf{X}}_{i,t}$ and $\underline{\mathbf{Y}}_{i,t}$ to the multiresolution fusion module to generate the initially fused MSIs $\underline{\mathbf{Z}}_{i,t}^{(\text{Fuse})}$;
5. Feed the last band of $\underline{\mathbf{Y}}_{i,t}$ to the eye-spiral feature extractor to generate the spiral feature maps $\mathbf{Y}_{i,f}^{(\text{Spiral-feature})}$ and the eye feature maps $\mathbf{Y}_{i,t}^{(\text{Eye-feature})}$ by Eqs. (2)-(3), and extract the textural features $\mathrm{t}\left(\mathbf{Y}_{l,t}^{(\text{Eye})}\right)$ by GLCM;
6. Inject $\mathbf{Y}_{i,f}^{(\text{Spiral-feature})}$ and $\mathbf{Y}_{i,t}^{(\text{Eye-feature})}$ to $\underline{\mathbf{Z}}^{(\text{Fuse})}$ to generate the enhanced structural features $\underline{\mathbf{Z}}_{i,t}$ via Eqs. (4)-(5);
7. Replace $\left\{\underline{\mathbf{X}}_{i,t}, \underline{\mathbf{Y}}_{i,t}\right\}$ with $\left\{\underline{\mathbf{X}}_{i,t-n}, \underline{\mathbf{Y}}_{i,t-n}\right\}$, repeat Steps 4-6 and obtain $\underline{\mathbf{Z}}_{i,t-n}$;
8. Feed $\underline{\mathbf{Z}}_{i,t}$ and $\underline{\mathbf{Z}}_{i,t-n}$ to the temporal-structural feature extractor to generate $\underline{\mathbf{Z}}^{(\text{Difference})}$ by Eq. (16);
   // MPDF
9. Fuse $\underline{\mathbf{Z}}_{i,t}, \underline{\mathbf{Z}}_{i,t-n}, \underline{\mathbf{Z}}_{i,t}^{(\text{Differese})}$ by Eq. (17) and extract the deep temporal image features $\mathbf{z}_{i,t+m}$;
10. Fuse $\mathrm{z}_{i,t+m}$, $\mathrm{s}\left(\underline{\mathbf{X}}_{i,t}\right)$, $\mathrm{p}\left(\underline{\mathbf{X}}_{i,t}\right)$ and $\mathrm{t}\left(\mathbf{Y}_{i,t}^{(\text{Eye})}\right)$;
    // Prediction and optimization
11. Predict the MSW speed $\hat{\mathrm{o}}_{i,t+m}$;
12. Compute the overall loss ℓ_1 by Eqs. (7), (9), (19) and (20);
13. Update Θ_1 by the Adam optimizer;
14. **End for**
15. **End While**

Return Θ_1 and $\left\{\hat{\mathrm{o}}_{i,t+m}\right\}_{i=1,t=n}^{I,T-m}$.

that, we merge $\underline{\mathbf{Z}}_{i,t}^{(\text{Feature-time})}$ and $\underline{\mathbf{Z}}_{i,t}^{(\text{Difference})}$, extracting the deep temporal image features $\mathrm{z}_{i,t+m}$ through a convolutional block and two fully connected layers.

Then, we calculate the translation speed s of TC in the same way as [5]

$$\mathrm{s} = [R \cdot \arccos(\sin(\mathrm{lat}_1)\sin(\mathrm{lat}_2) + \cos(\mathrm{lon}_1)\cos(\mathrm{lon}_2)\cos(\mathrm{lat}_1 - \mathrm{lat}_2))]/t \quad (18)$$

where R is the radius of the Earth with a mean value of 6370 km, t is the time taken by TC to move from point A to B, lon_1 and lat_1 are the longitude and the latitude of point A, respectively, and lon_2 and lat_2 are the longitude and the latitude of point B, respectively. Given the center latitude and longitude of each TC from the public track dataset, we obtain the current translation speed $\mathrm{s}(\underline{\mathbf{X}}_{i,t})$ of $\underline{\mathbf{X}}_{i,t}$ by (18). Then, we integrate the multimodal physical data $\mathrm{p}(\underline{\mathbf{X}}_{i,t})$, $\mathrm{s}(\underline{\mathbf{X}}_{i,t})$, and $\mathrm{t}(\mathbf{Y}_{i,t}^{(\text{Eye})})$ into the temporal image features $\mathrm{z}_{i,t+m}$ by two fully connected layers and a convolutional layer, finally predicting the MSW speed $\hat{\mathrm{o}}_{i,t+m}$ for the ith TC sample at the time $t+m$.

6) MSW Prediction and Optimization: The MSW speed prediction loss is measured by mean squared error (mse) between the predicted MSW speed $\hat{\mathrm{o}}_{i,t+m}$ and the real one $\mathrm{o}_{i,t+m}$ of the ith cyclone at the timestamp $t+m$

$$\ell^{(\text{Pre})} = \frac{1}{I(T-m-n)} \sum_{i=1}^{I} \sum_{t=n}^{T-m} \left(\mathrm{o}_{i,t+m} - \hat{\mathrm{o}}_{i,t+m}\right)^2 \quad (19)$$

where $m = 0, 6, 12, 18, 24$ h, and n is an empirically defined time lag that allows Pici-Net to exploit temporal features from historical samples. Given (7), (9), and (19), the total loss of Pici-Net is defined as

$$\ell_1 = \ell^{(\text{Pre})} + \lambda_1 \ell^{(\text{Eye})} + \lambda_2 \ell^{(\text{Fuse})} \quad (20)$$

where λ_1 and λ_2 are the tradeoff coefficients.

We train Pici-Net in an end-to-end fashion by the Adam optimizer with mini-batch gradient decent [47], as summarized in Algorithm 1. For simplicity, we denote Pici-Net as a nonlinear function $\psi_1(\bullet, \Theta_1)$, where Θ_1 holds all the learnable parameters.

B. Pici-Net+

Due to some TC datasets failing to provide multiresolution MSIs, we tailor the vanilla Pici-Net to the input of single-resolution MSIs, thus creating Pici-Net+. It mainly comprises TSFE, MPDF, and MSW prediction. Compared to Pici-Net, Pici-Net+ does not include the multiresolution fusion module, while keeping the eye-spiral pattern descriptor, a simplified version of the structural information injector, the temporal-structural feature extractor and the MPDF module from Pici-Net, as illustrated in Fig. 2. Similar to Pici-Net, Pici-Net+ obtains the spiral feature map $\mathbf{Y}_{i,t}^{(\text{Spiral-feature})} \in \mathbb{R}^{hw \times hw}$ via the self-attention block described by (2), the eye region feature map $\mathbf{Y}_{i,t}^{(\text{Eye-feature})} \in \mathbb{R}^{(hw/4) \times (hw/4)}$ from the last band of the input MSIs $\underline{\mathbf{X}}_{i,t} \in \mathbb{R}^{h \times w \times B}$ via the branch the same as Fig. 1(b) and the eye-region textural features $\mathrm{t}(\mathbf{Y}_{i,t}^{(\text{Eye})})$ via GLCM.

Meanwhile, there is some difference in the structural information injector. To inject the spiral structural information to the input MSIs, we fuse $\mathbf{Y}_{i,t}^{(\text{Spiral-feature})}$ and $\underline{\mathbf{X}}_{i,t}$ by

$$\underline{\mathbf{Z}}_{i,t}^{(\text{Fuse-spiral})} = \mathrm{reshape}\left(\mathbf{Y}_{i,t}^{(\text{Spiral-feature})} \cdot \mathbf{X}_{i,t}\right) \quad (21)$$

where $\mathbf{X}_{i,t} \in \mathbb{R}^{hw \times B}$ is reshaped from $\underline{\mathbf{X}}_{i,t}$. Then, we employ a local skip connection to reintroduce the original data and obtain the deep representations $\underline{\mathbf{Z}}_{i,t}^{(\text{Extract})}$ as follows:

$$\underline{\mathbf{Z}}_{i,t}^{(\text{Extract})} = \underline{\mathbf{Z}}_{i,t}^{(\text{Fuse-spiral})} + \underline{\mathbf{X}}_{i,t}. \quad (22)$$

To inject the eye-region information to the input MSIs, we draw the central part $\underline{\mathbf{Z}}_{i,t}^{(\text{Extract-central})} \in \mathbb{R}^{(h/2) \times (w/2) \times B}$ from $\underline{\mathbf{Z}}_{i,t}^{(\text{Extract})} \in \mathbb{R}^{h \times w \times B}$ and combine it with $\mathbf{Y}_{i,t}^{(\text{Eye-feature})} \in \mathbb{R}^{(hw/4) \times (hw/4)}$ output by (3) to obtain the fused eye features $\underline{\mathbf{Z}}_{i,t}^{(\text{Eye})} \in \mathbb{R}^{(h/2) \times (w/2) \times B}$ as follows:

$$\underline{\mathbf{Z}}_{i,t}^{(\text{Eye})} = \mathrm{reshape}\left(\mathbf{Y}_{i,t}^{(\text{Eye-feature})} \cdot \mathbf{Z}_{i,t}^{(\text{Extract-central})}\right) \quad (23)$$

where $\mathbf{Z}_{i,t}^{(\text{Extract-central})} \in \mathbb{R}^{(hw/4) \times B}$ is reshaped from $\underline{\mathbf{Z}}_{i,t}^{(\text{Extract-central})}$.

Comparing (23) to (5), we can see that the attention module is removed for simplicity. Then, we combine $\underline{\mathbf{Z}}_{i,t}^{(\text{Eye})}$ and the spiral portion of $\underline{\mathbf{Z}}_{i,t}^{(\text{Extract})}$ to produce $\underline{\mathbf{Z}}_{i,t}^{(\text{Fuse-eye})} \in \mathbb{R}^{h \times w \times B}$, add

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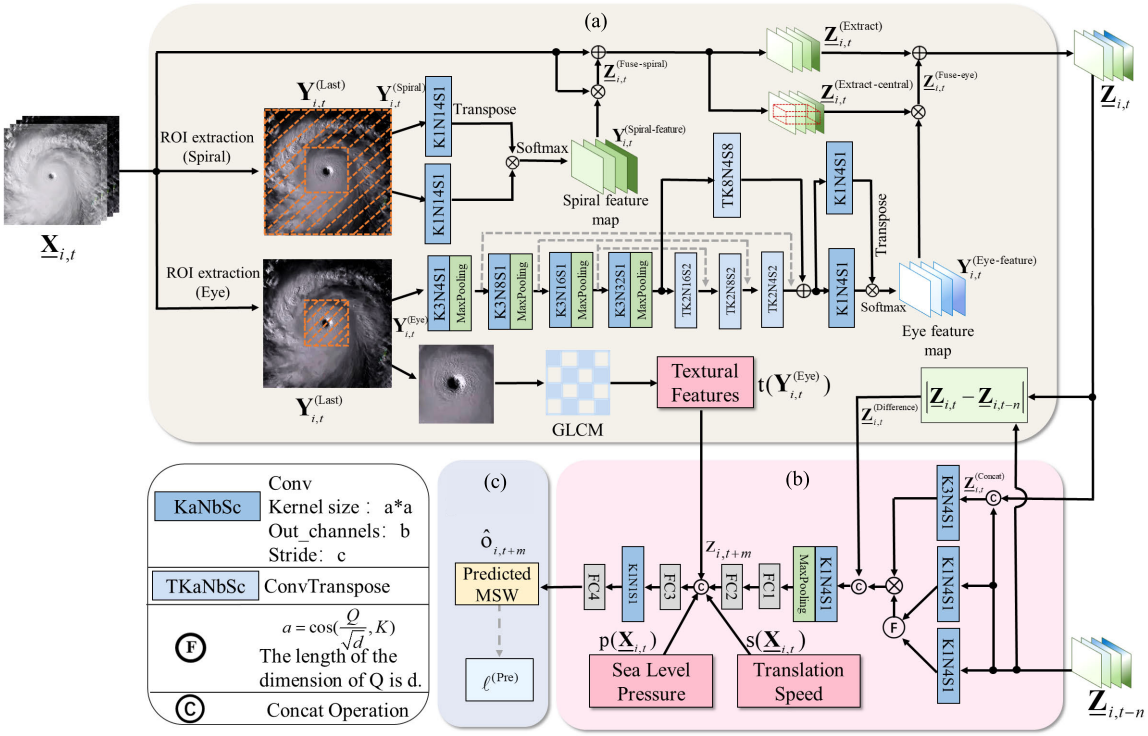


Fig. 2. Proposed Pici-Net+ for TC intensity prediction. (a) TSFE. (b) MPDF. (c) MSW prediction.

it to $\underline{\mathbf{Z}}_{i,t}^{(\text{Extract})}$ and obtain the enhanced structural features $\underline{\mathbf{Z}}_{i,t}$ guided by the TC structural knowledge. Similarly, we obtain the enhanced MSIs $\underline{\mathbf{Z}}_{i,t-n}$ at time $t-n$. After that, we extract temporal-structural features by differencing $\underline{\mathbf{Z}}_{i,t}$ and $\underline{\mathbf{Z}}_{i,t-n}$ as in Pici-Net described by (16).

The architecture of MPDF in Pici-Net+ is the same as that in Pici-Net. We feed $\underline{\mathbf{Z}}_{i,t}$, $\underline{\mathbf{Z}}_{i,t-n}$, $\mathrm{s}(\underline{\mathbf{X}}_{i,t})$, $\mathrm{p}(\underline{\mathbf{X}}_{i,t})$, and $\mathrm{t}(\mathbf{Y}_{i,t}^{(\text{Eye})})$ to MPDF and predict the MSW speed $\hat{\mathrm{o}}_{i,t+m}$, where $\mathrm{s}(\underline{\mathbf{X}}_{i,t})$ is the transitional speed computed by (18) and $\mathrm{p}(\underline{\mathbf{X}}_{i,t})$ is the sea level pressure provided by the track data. Given (19), the total loss of Pici-Net+ is defined as

$$\ell_2 = \frac{1}{I(T-m-n)} \sum_{i=1}^{I} \sum_{t=n}^{T-m} \left(\mathrm{o}_{i,t+m} - \hat{\mathrm{o}}_{i,t+m}\right)^2. \tag{24}$$

We train Pici-Net+ in an end-to-end fashion using the Adam optimizer with mini-batch gradient decent, as summarized in Algorithm 2. We denote Pici-Net+ as a nonlinear function $\psi_2(\bullet, \Theta_2)$, where Θ_2 holds all the learnable parameters.

C. Pici-Net++

Furthermore, we update Pici-Net+ and build Pici-Net++ to enhance the generalizability of the proposed networks. Pici-Net++ consists of two parts, a physics simulation model (PSM) to fill in the required but missing physical parameters and a prediction net (here Pici-Net+ is sued) to forecast TC MSW speed. Considering that there are TC datasets (e.g., HURSAT-B1) lacking the motional priors or the environmental factors, we derive PSM from Pici-Net+ as illustrated in Fig. 3 to simulate the parameters. Due to the efficacy of Pici-Net+ in our experiments, we use the model to build PSM. Denote the TC datasets with the required motional prior and the environmental factor as "source," and those without as "target." Input the source MSIs selected at the same wavelength as the target image and train PSM supervised by the motional prior and the environmental factor of the source MSIs. Afterward, input the target MSIs to the trained PSM and simulate the desired physical parameters. Feed the target MSIs and the simulated parameters to Pici-Net+ to complete multimodal fusion and MSW prediction as illustrated in Fig. 4. The details are as follows.

1) Physics Simulation Model: We modify Pici-Net+ to build PSM as illustrated in Fig. 3 to simulate the missing parameters in the dataset. Comparing Fig. 3 with Fig. 2, we can see that the major difference between PSM and Pici-Net+ lies in the outputs: PSM produces the TC translation speed and the minimum sea level pressure at the TC center, while Pici-Net+ predicts the MSW speed. To train PSM, we select the band images $\underline{\mathbf{X}}_{i,t}^{(\text{Source})}$ from the source data with the same wavelengths as the target MSIs $\underline{\mathbf{X}}_{i,t}^{(\text{Target})}$, since images of the same wavelength have similar physical properties. Then, we input $\underline{\mathbf{X}}_{i,t}^{(\text{Source})}$ with the real TC translation speed

$$\ell^{(\text{Simulation})} = \frac{\sum_{i=1}^{I} \sum_{t=n}^{T-m} \left|\mathrm{s}\left(\underline{\mathbf{X}}_{i,t}^{(\text{Source})}\right) - \hat{\mathrm{s}}\left(\underline{\mathbf{X}}_{i,t}^{(\text{Source})}\right)\right| + \left|\mathrm{p}\left(\underline{\mathbf{X}}_{i,t}^{(\text{Source})}\right) - \hat{\mathrm{p}}\left(\underline{\mathbf{X}}_{i,t}^{(\text{Source})}\right)\right|}{I(T-m-n)}. \tag{25}$$

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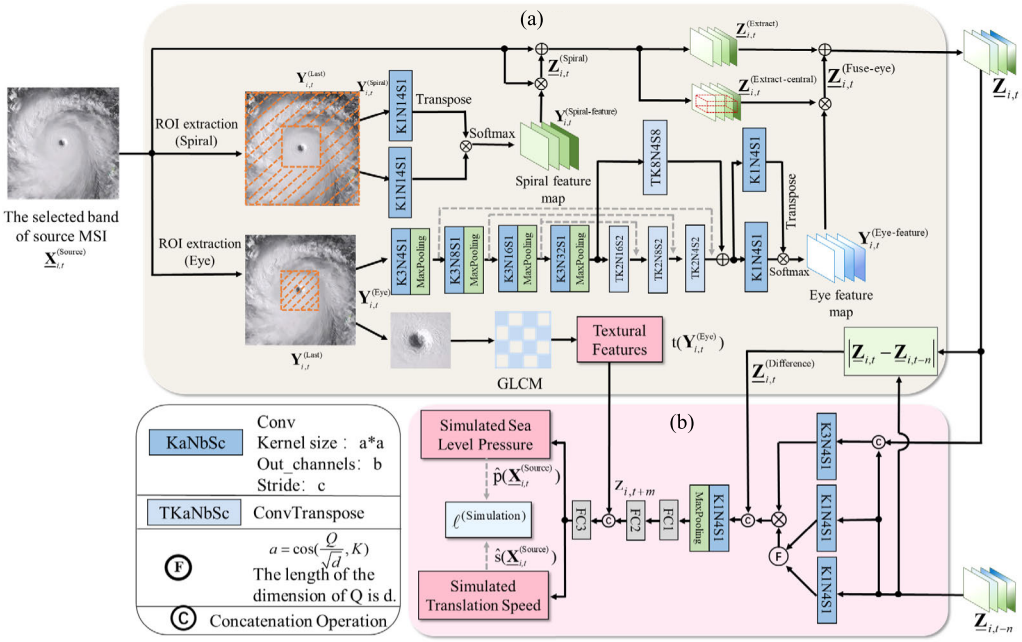


Fig. 3. Proposed PSM. (a) TSFE. (b) MPDF.

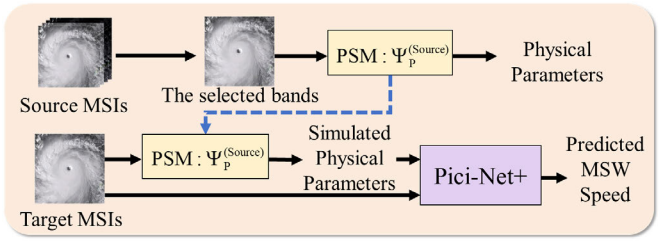


Fig. 4. Proposed Pici-Net++ for TC intensity prediction.

$\mathrm{s}(\underline{\mathbf{X}}_{i,t}^{(\mathrm{Source})})$ and the sea level pressure $\mathrm{p}(\underline{\mathbf{X}}_{i,t}^{(\mathrm{Source})})$ of $\underline{\mathbf{X}}_{i,t}^{(\mathrm{Source})}$ used as regression labels to train PSM and infer the simulated TC translation speed $\hat{\mathrm{s}}(\underline{\mathbf{X}}_{i,t}^{(\mathrm{Source})})$ and the simulated sea level pressure $\hat{\mathrm{p}}(\underline{\mathbf{X}}_{i,t}^{(\mathrm{Source})})$. The loss is measured by (25), as shown at the bottom of page 9.

We denote the PSM trained on the source data as $\Psi_{\mathrm{P}}^{(\mathrm{Source})}(\bullet, \Theta_{\mathrm{P}})$, where Θ_{P} holds all the learnable parameters. Therefore,

$$\left\{\hat{\mathrm{s}}\left(\underline{\mathbf{X}}_{i,t}^{(\mathrm{Source})}\right), \hat{\mathrm{p}}\left(\underline{\mathbf{X}}_{i,t}^{(\mathrm{Source})}\right)\right\} = \Psi_{\mathrm{P}}^{(\mathrm{Source})}\left(\underline{\mathbf{X}}_{i,t}^{(\mathrm{Source})}, \underline{\mathbf{X}}_{i,t-n}^{(\mathrm{Source})}, \Theta_{\mathrm{P}}\right). \tag{26}$$

Next, we input the target MSIs $\underline{\mathbf{X}}_{i,t}^{(\mathrm{Target})}$ to $\Psi_{\mathrm{P}}^{(\mathrm{Source})}$ and output the simulated parameters of $\underline{\mathbf{X}}_{i,t}^{(\mathrm{Target})}$ as follows:

$$\left\{\hat{\mathrm{s}}\left(\underline{\mathbf{X}}_{i,t}^{(\mathrm{Target})}\right), \hat{\mathrm{p}}\left(\underline{\mathbf{X}}_{i,t}^{(\mathrm{Target})}\right)\right\} = \Psi_{\mathrm{P}}^{(\mathrm{Source})}\left(\underline{\mathbf{X}}_{i,t}^{(\mathrm{Target})}, \underline{\mathbf{X}}_{i,t-n}^{(\mathrm{Target})}, \Theta_{\mathrm{P}}\right). \tag{27}$$

Finally, we feed the simulated parameters and the target MSIs to Pici-Net+ to predict the MSW speed $\hat{\mathrm{o}}_{i,t+m}^{(\mathrm{Target})}$.

2) Optimization: To save memory and improve prediction accuracy, we break down the training process of Pici-Net++ into two stages, first training PSM $\Psi_{\mathrm{P}}^{(\mathrm{Source})}(\bullet, \Theta_{\mathrm{P}})$ then Pici-Net $+\psi_2(\bullet, \Theta_2)$ as defined in Section IV-B. Here, the input to $\psi_2(\bullet, \Theta_2)$ consists of the TC MSIs $\underline{\mathbf{X}}_{i,t}^{(\mathrm{Target})}$ from the "target", the simulated parameters $\left\{\hat{\mathrm{s}}(\underline{\mathbf{X}}_{i,t}^{(\mathrm{Target})}), \hat{\mathrm{p}}(\underline{\mathbf{X}}_{i,t}^{(\mathrm{Target})})\right\}$ and the

Algorithm 2 Optimization of Pici-Net+

Input: the TC MSIs, the TC translation speed, the minimum sea level pressure at the TC center and the MSW speed $\left\{\underline{\mathbf{X}}_{i,t}, \underline{\mathbf{X}}_{i,t-m}, \mathrm{s}\left(\underline{\mathbf{X}}_{i,t}\right), \mathrm{p}\left(\underline{\mathbf{X}}_{i,t}\right), \mathrm{o}_{i,t+m}\right\}_{i=1,t=n}^{I,T-m}$;
Output: model $\psi_2(\cdot, \Theta_2)$ and the predicted MSW speed $\left\{\hat{\mathrm{o}}_{i,t+m}\right\}_{i=1,t=n}^{I,T-m}$;
Steps:

1. Initialize Θ_2;
2. **While** Θ_2 is not converged **do**
3. **for** $i = 1, 2, \ldots, I, t = n, n+1, \ldots, T-m$
   //TSFE
4. Feed $\underline{\mathbf{x}}_{i,t}$ and $\underline{\mathbf{x}}_{i,t-n}$ to TSFE to generate the structurally enhanced high-quality $\underline{\mathbf{Z}}_{i,t}$ and $\underline{\mathbf{Z}}_{i,t-n}$ via Eqs. (21)-(23);
5. Calculate the textural features $\mathrm{t}\left(\mathbf{Y}_{i,t}^{(\mathrm{Eye})}\right)$ by GLCM;
6. Extract the temporal-structural features $\underline{\mathbf{Z}}_{i,t}^{(\mathrm{Difference})}$ by Eq. (16);
   // MPDF
7. Fuse the multimodal data $\underline{\mathbf{Z}}_{i,t}, \underline{\mathbf{Z}}_{i,t-n}, \underline{\mathbf{Z}}_{i,t}^{(\mathrm{Difference})}, \mathrm{s}\left(\underline{\mathbf{X}}_{i,t}\right)$, $\mathrm{p}\left(\underline{\mathbf{X}}_{i,t}\right)$ and $\mathrm{t}\left(\mathbf{Y}_{i,t}^{(\mathrm{Eye})}\right)$;
   // Prediction and optimization
8. Predict the MSW speed $\hat{\mathrm{o}}_{i,t+m}$;
9. Compute the overall loss ℓ_2 by Eq. (24);
10. Update Θ_2 by Adam optimizer;
11. **End for**
12. **End while**
13. **Return** Θ_2 and $\left\{\hat{\mathrm{o}}_{i,t+m}\right\}_{i=1,t=n}^{I,T-m}$;

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MSW speed $o_{i,t+m}^{(\text{Target})}$ as labels, while the output is the predicted MSW speed $\hat{o}_{i,t+m}^{(\text{Target})}$ for the target data. The loss of $\psi_2(\bullet, \Theta_2)$ is ℓ_2 defined in (24). We train $\Psi_{\text{P}}^{(\text{Source})}(\bullet, \Theta_{\text{P}})$ and $\psi_2(\bullet, \Theta_2)$ separately by the Adam optimizer with mini-batch gradient decent. The training steps are summarized in Algorithm 3.

IV. Experiments

A. Dataset

For performance evaluation, we use three datasets: the self-collected dataset named FY4A-TC, the public dataset TCIR [15], and the public dataset HURSAT-B1 [16]. The details of the three datasets are shown in Table I.

FY4A-TC consists of 20 031 sets of multiresolution MSIs of 113 TCs over the Northwest Pacific captured by Chinese FY4A meteorological satellite [17] from 2018 to 2023. The multiresolution MSIs, i.e., HR-MSIs and LR-MSIs, are sampled by FY4A simultaneously in about 15-min intervals. With 14 spectral bands and a spatial resolution of 4 km, each set of LR-MSIs are cropped into a $240 \times 240 \times 14$ data cube. Each set of HR-MSIs with seven spectral bands and a spatial resolution of 2 km are cropped into a $480 \times 480 \times 7$ data cube. It is guaranteed that there is one TC presented in the central part of each set of MSIs after data cropping. The FY4A-TC dataset is spilt into three subsets, 15 522 sets of MSIs of 84 TCs for training, 3066 sets of MSIs of 29 TCs for validation, and 3059 sets of MSIs of 14 TCs for test. The MSW speed (m/s), the sea level pressure at the TC center (hPa), and the translation speed computed with the geographical coordinates of each TC are obtained from the best-track data of the China Meteorological Administration (CMA) [48], [49].

In the TCIR dataset, each set of MSIs, sampled at a temporal resolution of 3 h, consists of four spectral bands and are cropped into a $64 \times 64 \times 2$ data cube as only the IR and PMW bands are used following the convention [15]. The dataset includes the MSW speed (m/s), the pressure at the TC center (hPa) and the geographical coordinates of each TC obtained from the International Best Track Archive for Climate Stewardship (IBTrACS) [50], [51], while the translation speed is computed with the coordinates. The dataset is split in the same way as reference [5], [6], with 59 837 sets of MSIs of 1097 TCs from 2003 to 2014 for training, 10 662 sets of MSIs of 188 TCs from 2015 to 2016 for validation and 4580 sets of MSIs of 94 TCs from 2017 for test.

As for the HURSAT-B1 dataset, each set of MSIs consists of two bands sampled at a temporal resolution of 3 h. Following the convention [16], only the IR band is used here. The dataset includes the MSW speed (m/s) obtained from IBTrACS [50], [51] but not the physical parameters required by Pici-Nets. For the sake of physical data simulation, the HURSAT-B1 images are cropped to 240×240 pixels to match the spatial size of LR-MSIs in the FYA4-TC dataset. The dataset is spilt into three subsets, with 55 113 sets of MSIs of 1265 TCs from 2001 to 2013 for training, 5105 sets of MSIs of 86 TCs in 2014 for validation, and 6624 sets of MSIs of 95 TCs in 2015 for test.

Apart from splitting the datasets by the years, we also split them by geographical regions to demonstrate the regional efficacy and generalizability of the proposed Pici-Nets. It is known there are six ocean basins on the Earth, the Western Pacific (WP), the Eastern Pacific (EP), the South Pacific (SP), the North Atlantic (NA), the South Indian Ocean (SI), and the North Indian Ocean (NI) as shown in Fig. 5. It can be figured out from the track data that TCIR covers four basins, WP, EP, SP, and NA, while HURSAT-B1 covers six basins, WP, EP, SP, NA, SI, and NI. Hence, we come up with two ways to split the datasets according to the ocean basins. First, we train, validate, and test on the TC MSIs in each basin. For TCIR, the TC MSIs of each basin in 2003–2014, 2015 and 2016, and 2017 are used for training, validation, and test, respectively. For TCIR, the TC MSIs of each basin in 2001–2013, 2014, and 2015 are used for training, validation, and test, respectively. Second, for TCIR or HURSAT-B1, we use the TC MSIs in each one of the basins for test and the rest for training. While the first experiment sheds light on the single-basin performance of Pici-Nets, the second one demonstrates the generalizability of Pici-Nets across different basins. Note that FY4A-TC is not involved in these experiments because the dataset mainly covers the WP basin.

Algorithm 3 Optimization of Pici-Net++

Input: the target MSIs with their MSW speeds, and the selected bands of the source MSIs with the TC translation speed and the minimum sea level pressure at the TC center $\left\{\underline{\mathbf{X}}_{i,t}^{(\text{Target})}, \underline{\mathbf{X}}_{i,t-n}^{(\text{Target})}, o_{i,t+m}^{(\text{Target})}, \underline{\mathbf{X}}_{i,t}^{(\text{Source})}, \underline{\mathbf{X}}_{i,t-n}^{(\text{Source})}, s(\underline{\mathbf{X}}_{i,t}^{(\text{Source})}), p(\underline{\mathbf{X}}_{i,t}^{(\text{Source})})\right\}_{i=1,t=n}^{I,T-m}$;
Output: the models $\Psi_r^{(\text{Source})}(\cdot, \Theta_{\text{p}})$ and $\psi_2(\cdot, \Theta_2)$, the simulated physical data $\left\{\hat{s}(\underline{\mathbf{X}}_{i,t}^{(\text{Target})}), \hat{p}(\underline{\mathbf{X}}_{i,t}^{(\text{Target})})\right\}_{i=1,t=n}^{I,T-m}$; and the predicted MSW speed $\left\{\hat{o}_{i,t+m}^{(\text{Target})}\right\}_{i=1,t=n}^{I,T-m}$;
Steps:

1. Initialize Θ_{P}, Θ_2;
   //Train PSM
2. **While** Θ_{p} is not converged **do**
3. **for** $i = 1, 2, \ldots, I, t = n, n+1, \ldots, T-m$
4. Feed $\underline{\mathbf{X}}_{i,t}^{(\text{Target})}, \underline{\mathbf{X}}_{i,t-n}^{(\text{Target})}$ to PSM and obtain $\hat{s}\left(\underline{\mathbf{X}}_{i,t}^{(\text{Source})}\right)$ and $\hat{p}\left(\underline{\mathbf{X}}_{i,t}^{(\text{Source})}\right)$ by Eq. (26);
5. Compute $\ell^{(\text{Simulation})}$ with $\left\{s(\underline{\mathbf{X}}_{i,t}^{(\text{Source})}), p(\underline{\mathbf{X}}_{i,t}^{(\text{Source})})\right\}$ by Eq. (25);
6. Update Θ_{p} by the Adam optimizer;
7. **End for**
8. **End while**
   //Infer by PSM
9. Input $\left\{\underline{\mathbf{X}}_{i,t}^{(\text{Target})}, \underline{\mathbf{X}}_{i,t-1}^{(\text{Target})}\right\}_{i=1,t=n}^{I,T-m}$ to $\Psi_r^{(\text{Source})}(\cdot, \Theta_{\text{p}})$ and obtain the simulated parameters $\hat{s}\left(\underline{\mathbf{X}}_{i,t}^{(\text{Source})}\right)\hat{p}\left(\underline{\mathbf{X}}_{i,t}^{(\text{Source})}\right)_{i=1,t=n}^{I,T-m}$;
   //Train Pici-Net+
10. **While** Θ_2 is not converged **do**
11. **for** $i = 1, 2, \ldots, I, t = n, n+1, \ldots, T-m$
12. Input $\left\{\underline{\mathbf{X}}_{i,t}^{(\text{Target})}, \underline{\mathbf{X}}_{it-n}^{(\text{Target})}, \hat{s}\left(\underline{\mathbf{X}}_{i,t}^{(\text{Target})}\right), \hat{p}\left(\underline{\mathbf{X}}_{i,t}^{(\text{Target})}\right), o_{i,t+m}^{(\text{Target})}\right\}$ to PiciNet+ defined in **Algorithm 2** to predict $\hat{o}_{i,t+m}^{(\text{Target})}$;
13. Compute the loss ℓ_2 by Eq. (24);
14. Update Θ_2 by Adam optimizer;
15. **End for**
16. **End while**
17. **Return** Θ_{P}, Θ_2 and $\left\{\hat{s}\left(\underline{\mathbf{X}}_{i,t}^{(\text{Target})}\right), \hat{p}\left(\underline{\mathbf{X}}_{i,t}^{(\text{Target})}\right)\right), \hat{o}_{i,t+m}^{(\text{Target})}\right\}_{i=1,t=n}^{I,T-m}$.

B. Setup

Pici-Net is applied to the FY4A-TC dataset with the multiresolution MSIs. Pici-Net+ is applied to the FY4A-TC

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TABLE I
DESCRIPTION OF THE DATASETS

Property \ Dataset	FY4A-TC		TCIR [15]	HURSAT-B1 [16]
	LR-MSIs	HR-MSIs		
Width(height)	240	480	201	301
Bands number	14	7	4	2
Spatial resolution	4km	2km	4km	8km
Time interval	~15min	~15min	~3h	~3h
Dataset size	21647	21647	75079	66842
Covered area	Northwest Pacific	Northwest Pacific	Worldwide	Worldwide

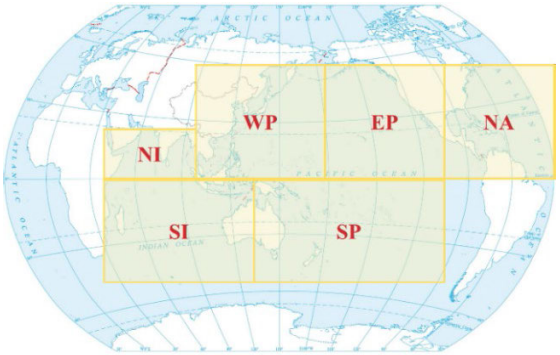


Fig. 5. Geographical distributions[1] of the ocean basins such as WP, EP, SP, NA, SI, and NI.

dataset and the TCIR dataset. Pici-Net++ is applied to the HURSAT-B1 dataset. Notably, we use PSM to simulate the missing parameters in HURSAT-B1, which, in this case, is regarded as the target data, with FY4A-TC leveraged as the source data, where the 13th band of each set of FY4A MSIs is selected to match the wavelength of the target data. The proposed Pici-Nets are compared with eight SOTA models, Deep-PHURIE [4], CNN-TC [3], CPT [37], Sharpen-CPT [37], TCICENet [5], T-TCNN [6], STD-AN [7], and MAF [13].

In the architecture ablation study, there are certain modules removed from Pici-Nets, denoted as "w/o spiral extractor," "w/o eye extractor," and "w/o MPDF." Moreover, we replace the self-attention block in the structural-information injector in Fig. 1(c) with cross-attention (CA) [52] and denote the ablated model by "with CA," to demonstrate the efficacy of the architecture of Pici-Net. In addition, to justify the dual-stage training strategy for Pici-Net++, we train PSM and Pici-Net+ in Pici-Net++ jointly and denote the model as "single-stage." For the physical data ablation, we run Pici-Net++ without some of the physical parameters while still simulating and fusing the others. The ablated models are denoted as "w/o S," "w/o P," and "w/o S+P," where S and P stand for the TC translation speed and the sea level pressure at the TC center, respectively. Furthermore, to validate the efficacy of the simulated data provided by PSM, we compare PSM with a traditional statistical method [46] that estimates the sea level pressure by the current MSW speed as follows:

$$\mathrm{P_e}\left(\underline{\mathbf{X}}_{i,t}^{(\mathrm{Target})}\right) = 1010 - \left(\mathrm{o}_{i,t}^{(\mathrm{Target})} \Big/ 3.4\right)^{1.55} \tag{28}$$

where 1010, 3.4, and 1.55 are the parameters fit from a large number of TC samples [46]. For the sea level pressure, we use the value estimated by (28) to train the prediction net, Pici-Net+ in the pipeline of Pici-Net++, and denote this ablated model as "with $\mathrm{P_e}$." For the other physics data used by this model, we acquire them in the same way as Pici-Net++. The reliability of the method is analyzed in Section IV-C4.

To evaluate the performances of the models on TC intensity (MSW speed) prediction, we use the public best-track data as the reference and obtain the mean absolute error (MAE) and the root mse (RMSE) of the predicted MSW speed. We also compute the standard deviation (STD) of MAE or RMSE to show the stability of the models.

For all the datasets, min–max normalization is applied to the original MSIs and the MSW speed labels. To efficiently train Pici-Nets, the learning rate is initially set to 0.001 and decay with a factor of 0.5 when the MAE of the validation set reaches a plateau. The batch size is 24. In (20), λ_1 is 0.001 and λ_2 is 0.01. The time lag n is set empirically based on the sampling time and the prediction time m of each dataset. For FY4A-TC, when m is 0, 6, 12, 18, or 24 h, n is set to 0.5, 0.5, 1, 1.5, or 2 h, respectively. For TCIR and HURSAT-B1, when m is 0, 6, 12, 18, or 24 h, n is set to 3, 3, 6, 6, or 6 h, respectively. All the algorithms are implemented in PyTorch on NVIDIA GeForce RTX 3080 with 64-GB RAM.

C. Result

1) General Performance: We evaluate the general performance of the proposed Pici-Nets on three datasets, FY4A-TC, TCIR, and HURSAT-B1 from a number of aspects as follows.

a) MSW speed prediction: As shown in Tables II–III, both Pici-Net and Pici-Net+ yield low prediction RMSE and MAE for FY4A-TC and TCIR. The results indicate that Pici-Net and Pici-Net+ are able to preserve and utilize the temporal–spectral–spatial information in multiresolution MSIs and the eye-spiral patterns of TCs. Moreover, Pici-Net outperforms Pici-Net+ on FY4A-TC, verifying the contributions of the multiresolution fusion to the TC MSW speed prediction.

b) Physical simulation: As can be seen from Table IV, Pici-Net++ predicts MSW speed with low RMSE and MAE for the HURSAT-B1 dataset. Since the physical parameters needed for multimodal fusion are missing from HURSAT-B1

[1]National Platform for Common Geospatial Information Services: http://bzdt.ch.mnr.gov.cn/index.html

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TABLE II
MAE AND RMSE OF MSW (m/s) PREDICTED FOR THE FY4A-TC DATASET

Time / Method	0h		6h		12h		18h		24h	
	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
Deep Phrie [4]	4.864± 0.239	3.659± 0.122	5.028± 0.263	3.761± 0.192	6.418± 0.132	4.913± 0.038	8.439± 0.092	6.337± 0.056	10.08± 0.115	7.651± 0.076
CNN-TC [3]	5.159± 0.031	3.799± 0.077	5.842± 0.242	4.384± 0.176	6.818± 0.123	5.113± 0.022	8.671± 0.115	6.543± 0.098	10.04± 0.091	7.708± 0.105
CPT [37]	4.118± 0.047	3.089± 0.075	5.554± 0.054	4.106± 0.155	6.545± 0.027	4.826± 0.006	8.194± 0.031	6.448± 0.085	10.07± 0.027	7.661± 0.006
Sharpen-CPT [37]	4.026± 0.179	3.005± 0.051	5.155± 0.183	3.917± 0.149	7.201± 0.171	5.175± 0.205	8.409± 0.084	6.281± 0.074	10.11± 0.125	7.752± 0.087
TCICENet [5]	4.762± 0.149	3.594± 0.093	6.427± 0.329	5.064± 0.115	6.144± 0.015	4.499± 0.071	8.394± 0.411	6.337± 0.332	9.908± 0.113	7.653± 0.047
T-TCNN [6]	4.991± 0.285	3.725± 0.058	6.344± 0.042	4.804± 0.261	6.617± 0.018	5.086± 0.189	8.899± 0.137	6.793± 0.039	10.74± 0.128	7.979± 0.091
STD-AN [7]	3.983± 0.113	2.954± 0.044	5.629± 0.284	3.881± 0.196	6.493± 0.161	4.815± 0.169	8.373± 0.208	6.118± 0.182	9.612± 0.278	7.391± 0.208
MAF [13]	2.832± 0.441	2.275± 0.417	3.974± 0.262	3.014± 0.215	6.645± 0.297	5.055± 0.236	8.291± 0.714	6.272± 0.718	9.561± 0.246	7.343± 0.155
Pici-Net	**2.633± 0.002**	**1.948± 0.011**	**3.829± 0.133**	**2.849± 0.064**	**5.473± 0.054**	**4.212± 0.078**	**7.338± 0.045**	**5.531± 0.039**	**8.827± 0.058**	**6.708± 0.076**
Pici-Net+	3.193± 0.024	2.441± 0.037	4.195± 0.033	3.096± 0.015	6.255± 0.118	4.623± 0.022	8.136± 0.073	6.036± 0.039	9.426±0. 039	7.074± 0.047

TABLE III
MAE AND RMSE OF MSW (m/s) PREDICTED FOR THE TCIR DATASET

Time / Method	0h		6h		12h		18h		24h	
	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
Deep Phrie [4]	5.129± 0.225	3.891± 0.144	5.604± 0.185	4.231± 0.132	6.627± 0.025	4.759± 0.047	8.118± 0.073	5.844± 0.106	9.404± 0.115	6.846± 0.087
CNN-TC [3]	4.917± 0.039	3.709± 0.026	5.491± 0.094	4.131± 0.047	6.521± 0.061	4.741± 0.053	8.051± 0.117	5.739± 0.111	9.273± 0.324	6.724± 0.236
CPT [37]	4.821± 0.018	3.672± 0.024	6.481± 0.055	4.871± 0.036	7.205± 0.037	5.301± 0.041	8.276± 0.029	5.947± 0.041	9.428± 0.051	6.886± 0.024
Sharpen-CPT [37]	4.758± 0.218	3.661± 0.175	6.605± 0.122	4.989± 0.092	7.272± 0.167	5.274± 0.121	8.552± 0.121	6.166± 0.134	9.478± 0.085	6.806± 0.068
TCICENet [5]	4.873± 0.158	3.735± 0.142	5.569± 0.146	4.112± 0.131	6.791± 0.026	4.891± 0.471	8.552± 0.034	5.965± 0.019	10.33±0. 106	7.368± 0.097
T-TCNN [6]	5.383± 0.211	4.186± 0.178	6.324± 0.024	4.791± 0.061	7.592± 0.254	5.484± 0.081	8.598± 0.068	5.964± 0.374	9.994± 0.135	7.159± 0.002
STD-AN [7]	4.735± 0.016	3.651± 0.009	4.981± 0.012	3.841± 0.035	6.536± 0.078	4.758± 0.029	7.861± 0.036	5.618± 0.052	9.344± 0.032	6.831± 0.094
MAF [13]	**2.664± 0.049**	**1.874± 0.037**	4.825± 0.173	3.670± 0.195	6.507± 0.058	4.642± 0.099	8.237± 0.102	5.843± 0.111	9.188± 0.079	6.531± 0.053
Pici-Net+	3.787± 0.012	2.983± 0.019	**4.728± 0.009**	**3.549± 0.032**	**6.258± 0.029**	**4.472± 0.016**	**7.775± 0.014**	**5.498± 0.036**	**9.054± 0.023**	**6.487± 0.028**

but simulated by PSM in Pici-Net++, the results verify the feasibility of our simulation idea, the efficacy of the proposed PSM and the usefulness of the simulated data for TC MSW speed prediction. For most multimodal-fusion-based models for TC intensity prediction, such as MAF [13], the reliance on the complete observational data prevents their being directly

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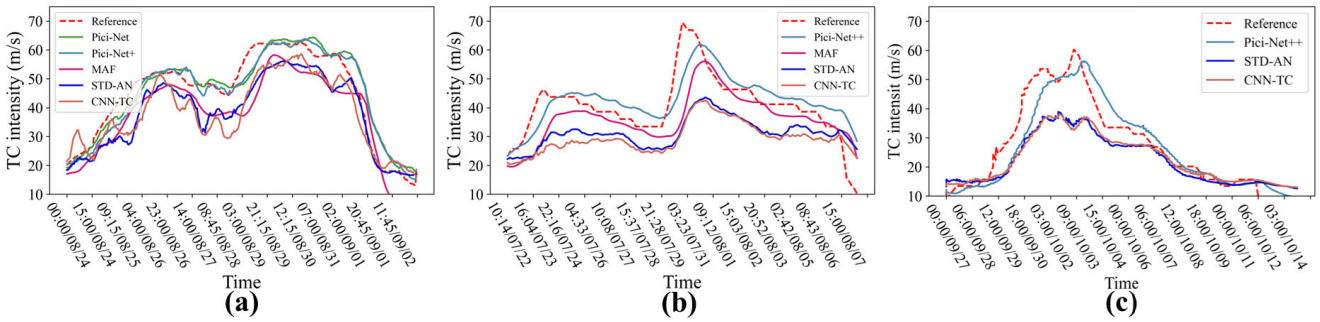


Fig. 6. TC intensity predicted by Pici-Nets and three well-performing models, MAF, CNN-TC, and STD-AN in 6 h for (a) TC SAOLA of the FY4A-TC dataset, (b) WPAC201707W of the TCIR dataset, and (c) Joaquin of the HURSAT-B1 dataset.

TABLE IV
MAE AND RMSE OF MSW (m/s) PREDICTED FOR THE HURSAT-B1 DATASET

Method \ Time	0h		6h		12h		18h		24h	
	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
Deep Phrie [4]	4.484± 0.116	3.269± 0.101	4.983± 0.032	3.625± 0.009	5.773± 0.041	4.071± 0.057	6.874± 0.008	4.809± 0.058	7.999± 0.105	5.638± 0.044
CNN-TC [3]	4.785± 0.032	3.436± 0.021	4.849± 0.035	3.465± 0.121	5.643± 0.071	4.138± 0.054	6.905± 0.046	4.788± 0.034	8.156± 0.099	5.591± 0.075
CPT [37]	4.213± 0.028	3.109± 0.038	5.541± 0.027	4.589± 0.024	6.178± 0.066	4.306± 0.125	7.081± 0.021	4.963± 0.062	8.071± 0.045	5.688± 0.056
Sharpen-CPT [37]	4.656± 0.085	3.417± 0.023	5.835± 0.065	4.708± 0.081	6.292± 0.073	4.491± 0.123	7.088± 0.081	4.992± 0.079	8.065± 0.087	5.687± 0.072
TCICENet [5]	5.191± 0.092	3.764± 0.061	5.289± 0.131	3.744± 0.121	6.228± 0.076	4.335± 0.049	7.196± 0.122	4.951± 0.069	8.499± 0.173	5.795± 0.125
T-TCNN [6]	5.581± 0.211	4.164± 0.178	6.497± 0.228	4.327± 0.158	7.923± 0.141	5.251± 0.053	8.961± 0.276	5.905± 0.016	9.217± 0.183	6.299± 0.067
STD-AN [7]	4.351± 0.014	3.246± 0.007	4.735± 0.048	3.443± 0.053	5.588± 0.011	4.016± 0.021	7.013± 0.035	4.963± 0.045	8.006± 0.014	5.794± 0.015
Pici-Net++	**2.987± 0.021**	**2.449± 0.021**	**3.907± 0.033**	**2.847± 0.044**	**5.159± 0.022**	**3.445± 0.059**	**6.731± 0.049**	**4.653± 0.039**	**7.821± 0.026**	**5.508± 0.059**

applied to the incomplete datasets, which, however, are quite common in the real world. In this work, with the introduction of PSM, the data dependency is largely reduced, allowing the proposed Pici-Nets to perform well even in the absence of partial data. In Section IV-C4, we will further analyze the contribution of each simulated parameter to the prediction ability of Pici-Net++ through the ablation experiments.

c) Tracking sensibility: In Fig. 6, we draw the curves of TC intensity predicted six hours ahead of the current time by Pici-Net and the competing models, MAF, CNN-TC, and STD-AN, along with the reference curves for three super TCs from different datasets. It can be seen that as TCs rapidly intensify, even surpassing 51 m/s, the predictions by Pici-Nets are the closest to the reference. It indicates that Pici-Nets exhibit good tracking sensibility. This promising result is partly attributed to the TSFE module of Pici-Nets that learns the temporal structural difference between the current and the past MSIs, and partly attributed to the MPDF module that leverages the physical parameters closely related to the current MSW speed.

d) Comparison with SOTA: Tables II–IV show that the proposed Pici-Nets outperform the competing models in TC intensity prediction. As can be seen in Table II, for the FY4A-TC dataset with multiresolution MSIs, when the time lapse is 0h, Pici-Net exceeds the best-competing method by 7.4% in terms of MAE; this gap amounts to 9.47% when the time-lapse prolongs to 24 h. For HURSAT-B1 without complete physical parameters, Table IV shows that Pici-Net++ exceeds the best-competing method, CPT by 3.26%–61.2% in terms of MAE as the time-lapse prolongs from 0 to 24 h. As most of the existing TC MSW speed estimation or prediction models are designed for IR images, they can only use spatial features of the satellite images, lacking information from any different perspective, thus not performing well in TC intensity prediction with MSIs. Meanwhile, the information leveraged by Pici-Nets includes the temporal–spectral–spatial features in the multiresolution MSIs and the dynamic physical data related to TC evolvement, which gives the proposed models a comprehensive understanding of TCs, naturally resulting in better predictions.

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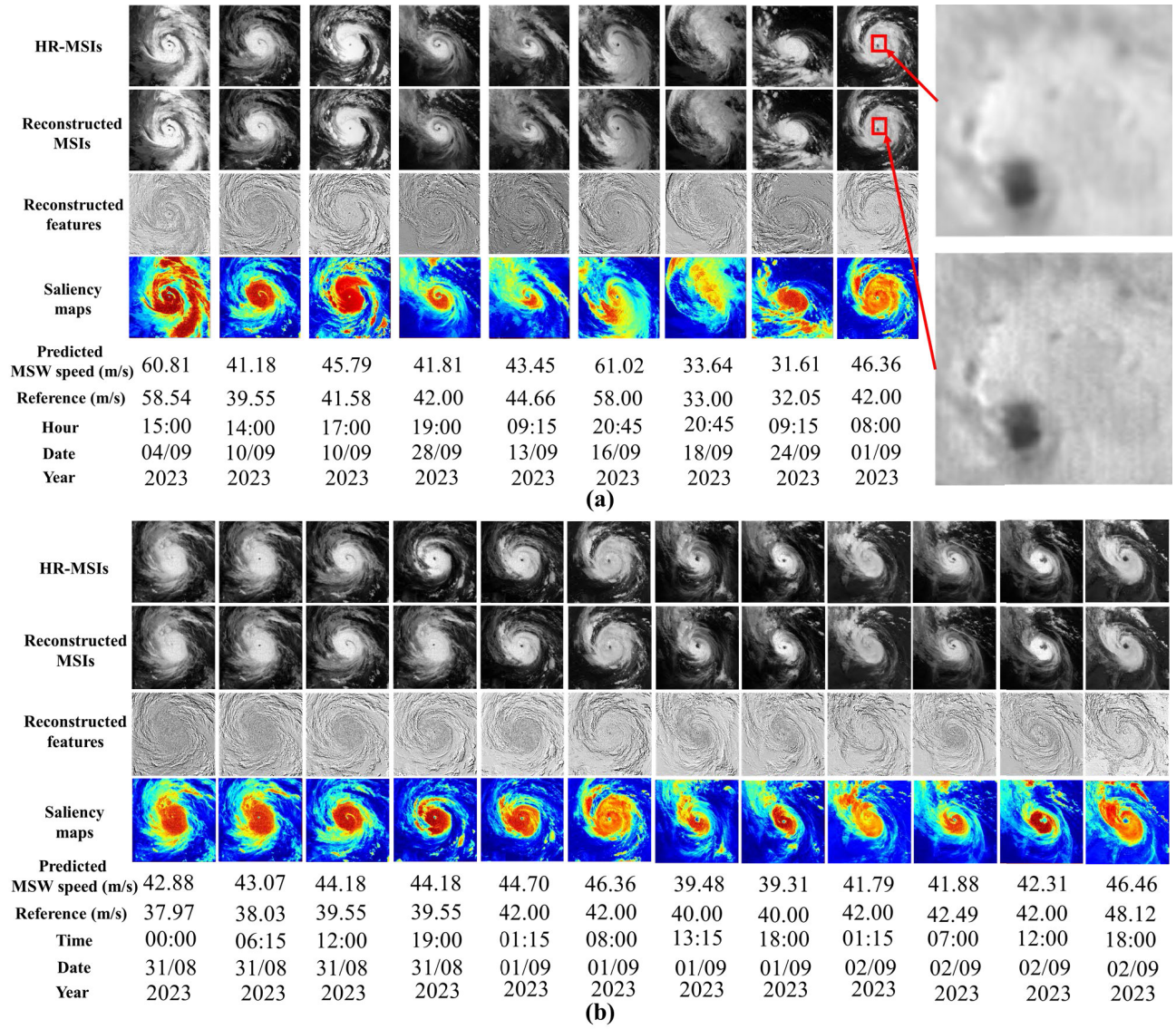


Fig. 7. Intermediate feature maps generated by Pici-Net for (a) different TC samples and (b) TC "SAOLA" samples at variable time stamps from the FY4A-TC dataset. The first and the second rows display the 7th band of the original HR-MSIs $\underline{\mathbf{Y}}_{i,t}$ and the reconstructed MSIs $\underline{\mathbf{Z}}_{i,t}$, respectively. The third row visualizes the reconstructed features (the whiter, the larger, and the difference), which represent the difference between original HR-MSIs and reconstructed MSIs. The fourth row shows the attention saliency maps (the redder and the larger saliency) of $\underline{\mathbf{Z}}_{i,t}^{(\text{Feature})}$ generated by structural information injector illustrated by Fig. 1(c). Underneath are the predicted and the reference MSW speed (m/s) along with the time stamps.

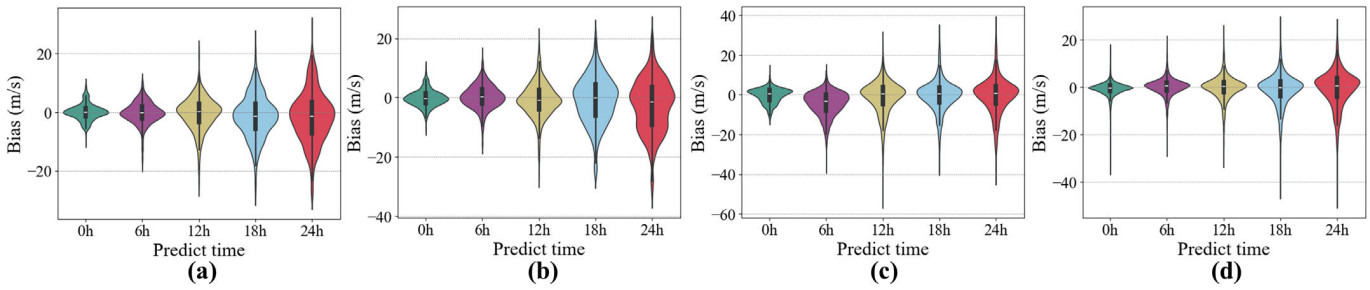


Fig. 8. Violin plots of the sample-wise bias of the MSW speed predicted by (a) Pici-Net on FY4A-TC, (b) Pici-Net+ on LR-MSIs of FY4A-TC, (c) Pici-Net+ on TCIR, and (d) Pici-Net++ on HURSAT-B1. Each violin represents the distribution of the sample-wise bias of a dataset while the white dot in each violin represents the median of the bias values.

Although some of the competing models use multiple types of information, they are limited in many ways. For instance, T-TCNN and TCICENET perform intensity categorization (classification) and estimation (regression) in separate stages, where errors of misclassification are inevitably passed on to the successive regressor. CPT and Sharpened-CPT take into

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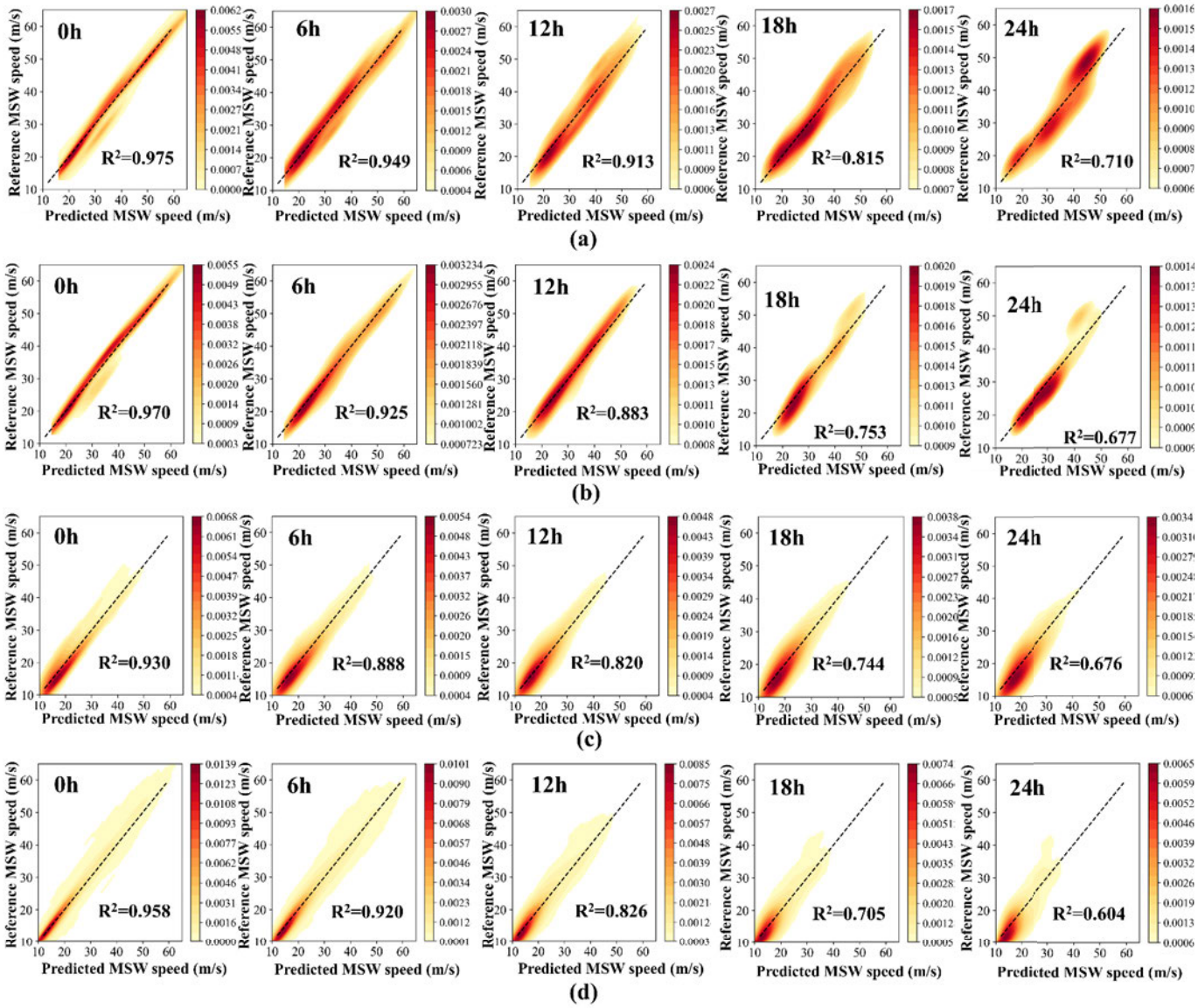


Fig. 9. Joint probability density distribution plots of the reference and the MSW speed predicted by (a) Pici-Net on FY4A-TC, (b) Pici-Net+ on LR-MSIs of FY4A-TC, (c) Pici-Net+ on TCIR, and (d) Pici-Net++ on HURSAT-B1. The coefficient of determination (R^2) for the predicted MSW speed and the reference is given in each plot. The colors painted along the diagonal represent the occurrence frequency obtained by kernel density estimation.

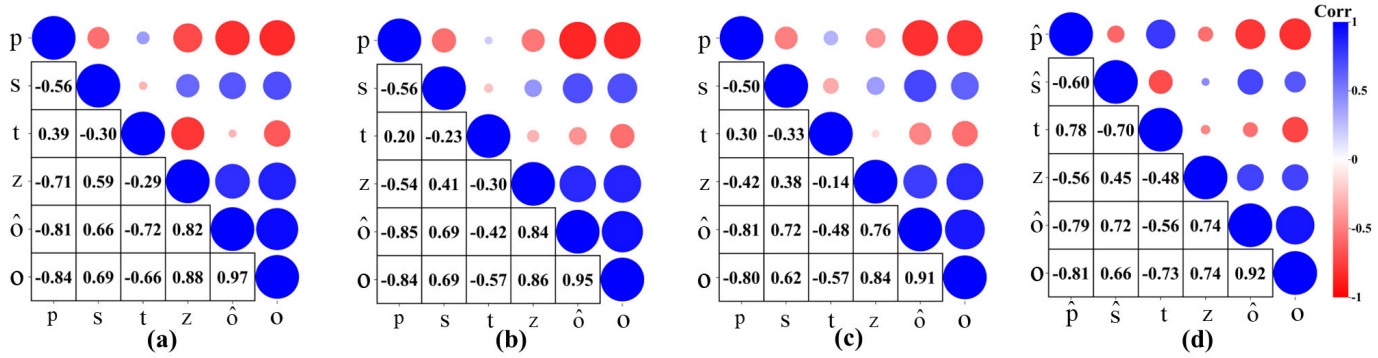


Fig. 10. Correlation heatmaps of sea level pressure, TC translation speed, textural feature, deep temporal image features before multimodal fusion, predicted MSW speed, and reference MSW speed used by (a) Pici-Net for FY4A-TC, (b) Pici-Net+ for LR-MSIs of FY4A-TC, (c) Pici-Net+ for TCIR, and (d) Pici-Net++ on HURSAT-B1. The colored circles stand for the absolute correlation coefficients. The blue circles represent positive correlation and the red ones represent negative correlation.

account the spatial information of MSIs, but they neglect that TC is a temporal system, where the current status of TCs can be influenced by the past. STD-AN extracts the temporal–spectral–spatial features from MSIs but does not consider the

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TABLE V
RMSE AND MAE OF MSW (m/s) PREDICTED BY DIFFERENT ABLATED MODELS FOR THE FY4A-TC DATASET

Method \ Time	0h		6h		12h		18h		24h	
	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
w/o spiral extractor	3.740± 0.131	2.925± 0.065	4.169± 0.054	3.131± 0.052	6.435± 0.092	4.672± 0.033	8.364± 0.062	6.442± 0.055	9.501± 0.025	7.429± 0.021
w/o eye extractor	3.926± 0.157	3.043± 0.077	4.394± 0.021	3.251± 0.013	6.679± 0.081	5.035± 0.029	9.048± 0.067	7.477± 0.049	9.762± 0.035	7.465± 0.075
w/o MPDF	5.182± 0.532	4.114± 0.328	5.963± 0.764	4.297± 0.437	7.521± 0.711	5.493± 0.435	10.02± 0.562	7.367± 0.385	12.12± 0.925	8.764± 0.721
with CA	3.825± 0.056	2.874± 0.036	5.451± 0.119	4.182± 0.176	6.802± 0.488	4.939± 0.368	8.368± 0.635	6.298± 0.484	10.23± 1.469	7.597± 0.873
w/o S	3.682± 0.052	3.022± 0.084	4.681± 0.179	3.574± 0.229	6.368± 0.044	4.834± 0.086	8.372± 0.191	6.341± 0.167	9.772± 0.023	5.688± 0.056
w/o P	3.683± 0.052	3.022± 0.084	5.543± 0.549	4.198± 0.274	7.098± 0.099	5.365± 0.228	9.209± 0.117	7.111± 0.228	10.72± 0.109	5.687± 0.072
w/o S+P	4.928± 0.528	3.678± 0.295	5.842± 0.484	4.159± 0.375	7.696± 0.815	5.782± 0.609	9.814± 0.524	7.363± 0.462	11.34± 0.967	7.699± 0.801
Pici-Net	**2.633± 0.002**	**1.948± 0.011**	**3.829± 0.133**	**2.849± 0.064**	**5.473± 0.054**	**4.212± 0.078**	**7.338± 0.045**	**5.531± 0.039**	**8.827± 0.058**	**6.708± 0.076**

TABLE VI
RMSE AND MAE OF MSW (m/s) PREDICTED BY DIFFERENT ABLATED MODELS FOR THE FY4A-TC DATASET (LR-MSIs)

Method \ Time	0h		6h		12h		18h		24h	
	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
w/o spiral extractor	3.665± 0.092	2.866± 0.061	4.317± 0.128	3.119± 0.064	6.203± 0.064	4.881± 0.048	8.869± 0.058	6.768± 0.027	9.971± 0.074	7.389± 0.039
w/o eye extractor	3.419± 0.284	2.725± 0.303	4.507± 0.152	3.413± 0.131	6.381± 0.035	4.671± 0.029	9.043± 0.157	6.791± 0.089	10.26± 0.011	7.954± 0.007
w/o MPDF	6.381± 0.568	4.737± 0.542	6.844± 0.658	5.225± 0.459	7.273± 0.715	5.334± 0.437	9.925± 0.824	7.508± 0.729	11.34± 0.965	8.607± 0.848
w/o S	4.105± 0.062	3.203± 0.107	5.027± 0.211	3.654± 0.194	6.633± 0.114	4.849± 0.089	8.996± 0.023	6.879± 0.021	10.46± 0.006	8.013± 0.009
w/o P	5.621± 0.142	4.228± 0.137	5.754± 0.097	4.289± 0.077	6.943± 0.043	5.182± 0.031	9.537± 0.235	7.273± 0.211	10.55± 0.135	8.357± 0.146
w/o S+P	5.778± 0.159	4.209± 0.092	6.622± 0.132	5.202± 0.078	7.201± 0.286	5.171± 0.346	9.633± 0.392	7.271± 0.189	11.03± 0.552	8.452± 0.495
Pici-Net+	**3.193± 0.024**	**2.441± 0.037**	**4.195± 0.033**	**3.096± 0.015**	**6.255± 0.118**	**4.623± 0.022**	**8.136± 0.073**	**6.036± 0.039**	**9.426± 0.039**	**7.074± 0.047**

physical parameters that reflect the dynamic status of TCs, thus being less sensitive to temporal changes in TC MSW than Pici-Nets, as reflected by Fig. 6. Interestingly, MAF achieves good performance in short-term TC intensity prediction, even surpassing Pici-Net+ when estimating the current intensity (i.e., the prediction time is set to 0 h), as indicated by Table III. It should be attributed to the fact that MAF incorporates multiple TC-related factors, including the current MSW speed as the observation data. However, as the time-lapse increases, Pici-Net+ surpasses MAF in prediction accuracy, since the observational data do not inform MAF of the temporal changes in TC structures. Moreover, while the real-world datasets, e.g., HURSAT-B1, are mostly short of the observational data required by MAF, the proposed PSM in Pici-Net++ can simulate the missing data, thereby alleviating the dependency of multimodal models on the specific datasets.

2) Explainability: The explainability of Pici-Net can be reflected by the temporal structural features learned by the model. To this end, we visualize the feature maps of different TC samples with the predicted and the reference MSW speed values in Fig. 7.

First, we demonstrate that Pici-Net can learn meaningful structural features for MSW speed prediction. It is known that stronger TCs usually exhibit clearer eye structures and

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TABLE VII
RMSE AND MAE OF MSW (m/s) PREDICTED BY DIFFERENT ABLATED MODELS FOR THE HURSAT-B1 DATASET

Method \ Time	0h		6h		12h		18h		24h	
	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
w/o spiral extractor	3.245± 0.085	2.237± 0.023	4.177± 0.052	2.974± 0.018	5.288± 0.101	3.547± 0.052	6.965± 0.029	4.835± 0.013	8.223± 0.094	5.624± 0.065
w/o eye extractor	3.275± 0.041	2.638± 0.094	4.33±0. 013	3.043± 0.083	5.465± 0.011	3.699± 0.052	7.268± 0.154	5.182± 0.108	8.271± 0.063	5.839± 0.101
w/o MPDF	5.302± 0.116	3.841± 0.101	5.726± 0.461	4.089± 0.351	6.327± 0.584	4.428± 0.496	9.401± 0.648	6.367± 0.539	11.57± 1.283	7.048± 0.942
single-stage	3.925± 0.121	2.987± 0.173	4.987± 0.147	3.797± 0.205	6.071± 0.291	4.453± 0.323	7.453± 0.189	5.263± 0.295	9.188± 0.798	6.431± 0.537
w/o S	4.304± 0.031	3.164± 0.077	4.427± 0.023	3.064± 0.116	5.646± 0.007	3.835± 0.013	7.559± 0.093	5.281± 0.105	8.921± 0.049	6.655± 0.177
w/o P	4.761± 0.025	3.591± 0.031	4.923± 0.288	3.395± 0.271	5.823± 0.057	4.119± 0.163	8.112± 0.057	5.703± 0.217	9.992± 0.045	7.422± 0.211
w/o S+P	4.829± 0.165	3.513± 0.098	5.113± 0.274	3.715± 0.168	5.977± 0.551	4.196± 0.426	8.394± 0.652	5.918± 0.446	10.35± 0.795	7.347± 0.698
with P_e	4.542± 0.135	3.559± 0.108	5.231± 0.092	4.109± 0.131	7.207± 0.058	5.982± 0.137	8.596± 0.027	6.082± 0.059	9.893± 0.058	7.023± 0.054
Pici-Net++	**2.987± 0.021**	**2.449± 0.021**	**3.907± 0.033**	**2.847± 0.044**	**5.159± 0.022**	**3.445± 0.059**	**6.742± 0.049**	**4.596± 0.039**	**7.821± 0.026**	**5.508± 0.059**

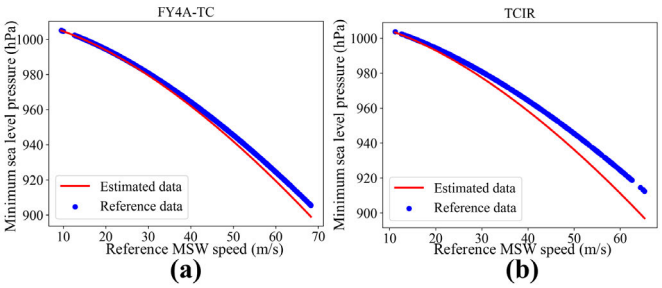


Fig. 11. Difference between the sea level pressure estimated by (28) and the reference value of the (a) FY4A-TC and (b) TCIR datasets.

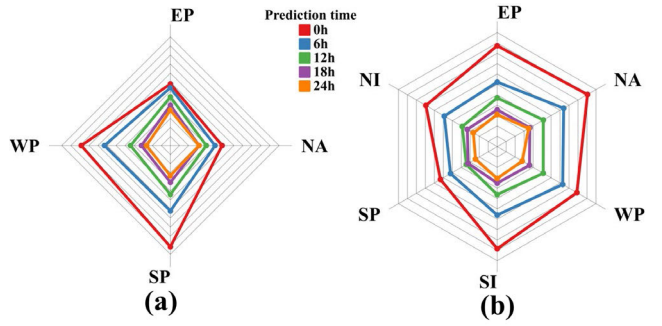


Fig. 12. MSW-speed prediction ability of (a) Pici-Net+ on TCIR and (b) Pici-Net++ on HURSAT-B1 in the single-basin experiment. A polygon visualizes the performance index derived from the reciprocal normalization of MAE of the MSW speed predicted in 0, 6, 12, 18, or 24 h. Each vertex of a polygon represents a basin. The farther a vertex is from the center, the stronger the predictive ability is.

denser spiral cloud bands [14], as illustrated by the first row of Fig. 7(a), suggesting the importance of the eye-spiral patterns to intensity prediction. The first and the second

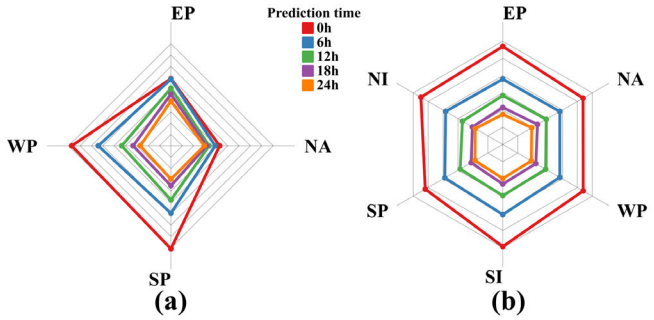


Fig. 13. MSW-speed prediction ability of (a) Pici-Net+ on TCIR and (b) Pici-Net++ on HURSAT-B1 in the cross-basin experiment. A polygon visualizes the performance index derived from the reciprocal normalization of MAE of the MSW speed predicted in 0, 6, 12, 18, or 24 h. Each vertex of a polygon represents a basin. The farther a vertex is from the center, the stronger the predictive ability is.

rows of Fig. 7(a), especially the zoomed-in view images on the right show that these crucial patterns in TC MSIs are enhanced by Pici-Net through multiresolution fusion and structural information injection. The reconstructed features of each TC sample visualized as grayscale maps in the third row indicate that Pici-Net can learn the overall structures of the TC samples while enhancing the textural details in the eye regions. The saliency maps in the fourth row show that the structural information injector as illustrated by Fig. 1(c) can drive the network to focus on the crucial patterns such as the cyclone eye and the cloud spiral for MSW speed prediction. The more parts of the meaningful cyclone structures highlighted by the salient area, the smaller the prediction error for the TC sample.

As for the evaluation of TSFE, we analyze the temporal feature maps learned by Pici-Net for a cyclone evolving over time in Fig. 7(b). From the third row, we can see that Pici-Net

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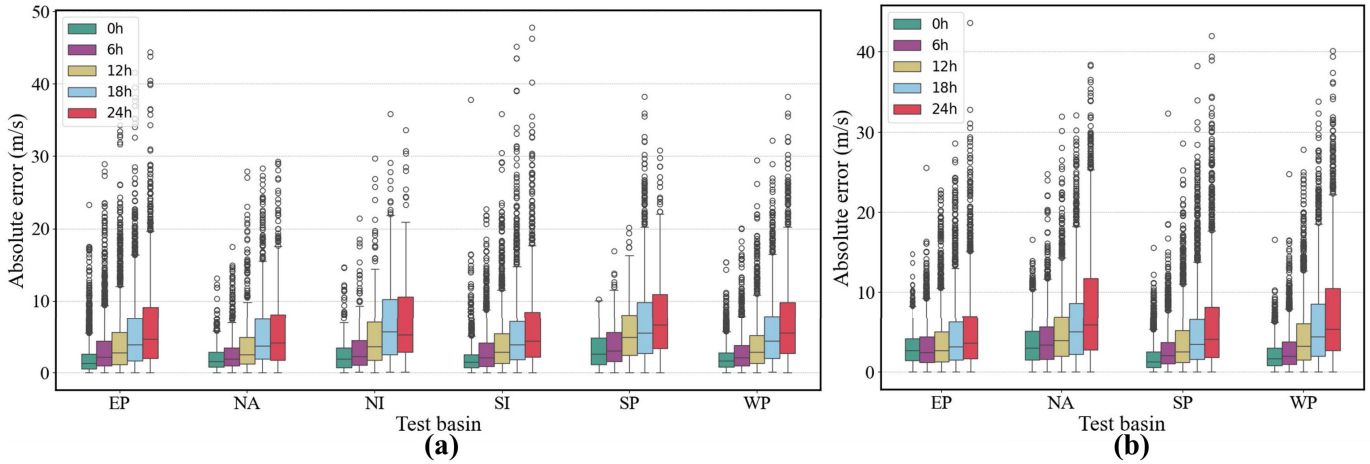


Fig. 14. Boxplot of absolute error over the prediction time for (a) Pici-Net+ on TCIR and (b) Pici-Net++ on HURSAT-B1 in the single-basin experiment. The boxes represent the distribution of the absolute errors and the black dashed line in each box represents the median of the absolute errors.

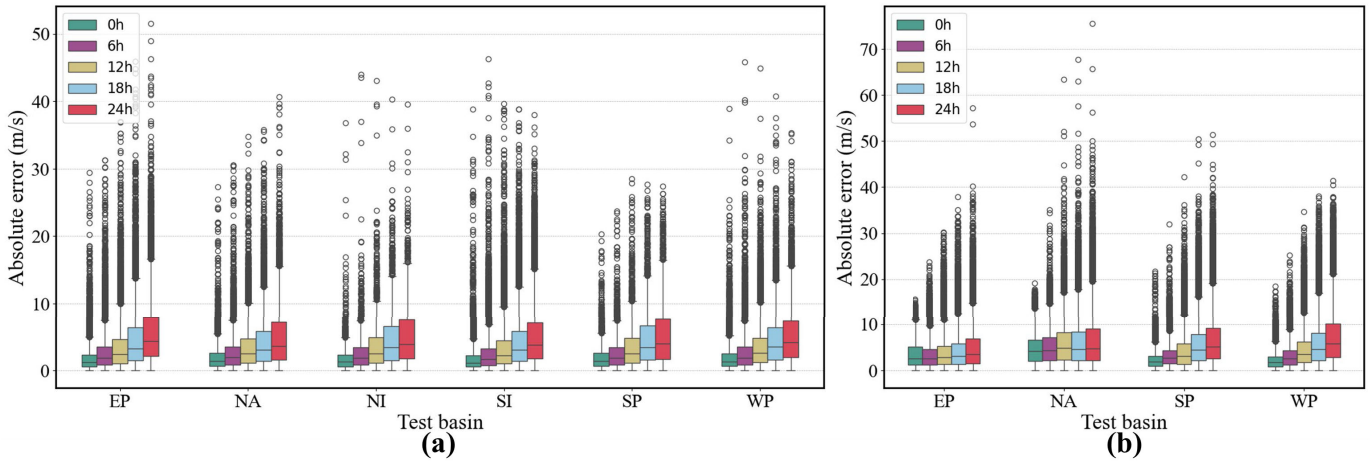


Fig. 15. Boxplot of absolute error over the prediction time for (a) Pici-Net+ on TCIR and (b) Pici-Net++ on HURSAT-B1 in the cross-basin experiment. The boxes represent the distribution of the absolute errors and the black dashed line in each box represents the median of the absolute errors.

enhances the details of the TC structures that change over the time in the fused high-quality MSIs. From the saliency maps in the fourth row, we can see that Pici-Net accurately captures the temporal cloud structures, reflecting the temporal changes in the eye region and the spiral rainbands that are crucial to TC MSW speed. These results indicate that the features learned by Pici-Nets are meaningful and the learning process can be explained as Pici-Nets incorporate the physical knowledge of TCs in feature extraction and intensity prediction.

3) Interpretability: To demonstrate the interpretability of the proposed models, we analyze the sample-wise results predicted by Pici-Nets with the bias (i.e., sample-wise error of the predicted MSW speed) over the variable time lap illustrated by Fig. 8 and the joint probability density distribution of the reference and predicted MSW speed in Fig. 9. As can be seen in Fig. 8, most median values of the bias are close to 0. As shown in Fig. 9, the majority of the MSW speed values predicted by the proposed model are close to the reference values, with the fitting coefficient R^2 close to 1. The results indicate that the Pici-Nets are robust to sample-wise variations, as they are guided by real or simulated physical parameters to effectively confine the prediction errors within small margins.

Furthermore, we interpret the multimodal fusion in Pici-Nets by analyzing the contribution of each modal of data (real or simulated) to MSW speed prediction. By this analysis, we can also interpret PSM and rationalize data simulation. In Fig. 10, we visualize the correlations between any two of the following variables, the sea level pressure "p" (or the simulated value "$\hat{p}$"), the TC translation speed "s" (or the simulated value "$\hat{s}$"), the structural features "t," the deep temporal image features "z" involved in multimodal fusion, the predicted MSW speed "$\hat{o}$" and the reference MSW speed "o." The colored circles stand for the absolute correlation coefficients. The larger the circle, the greater the correlation (no matter if it is positive or negative) [53]. As shown in Fig. 10(a)–(c), the correlation between the predicted MSW speed and the reference is higher than the correlation between the deep temporal image features and the reference. It suggests that the intensity prediction indeed benefited from the multimodal fusion. The translation speed reflects the distance traveled by a cyclone within a unit of time, while the sea level pressure reflects the environmental conditions during the occurrence of the TC. Fusing these parameters with the structural features and the deep temporal image features, Pici-Net and Pici-Net+ can imitate human

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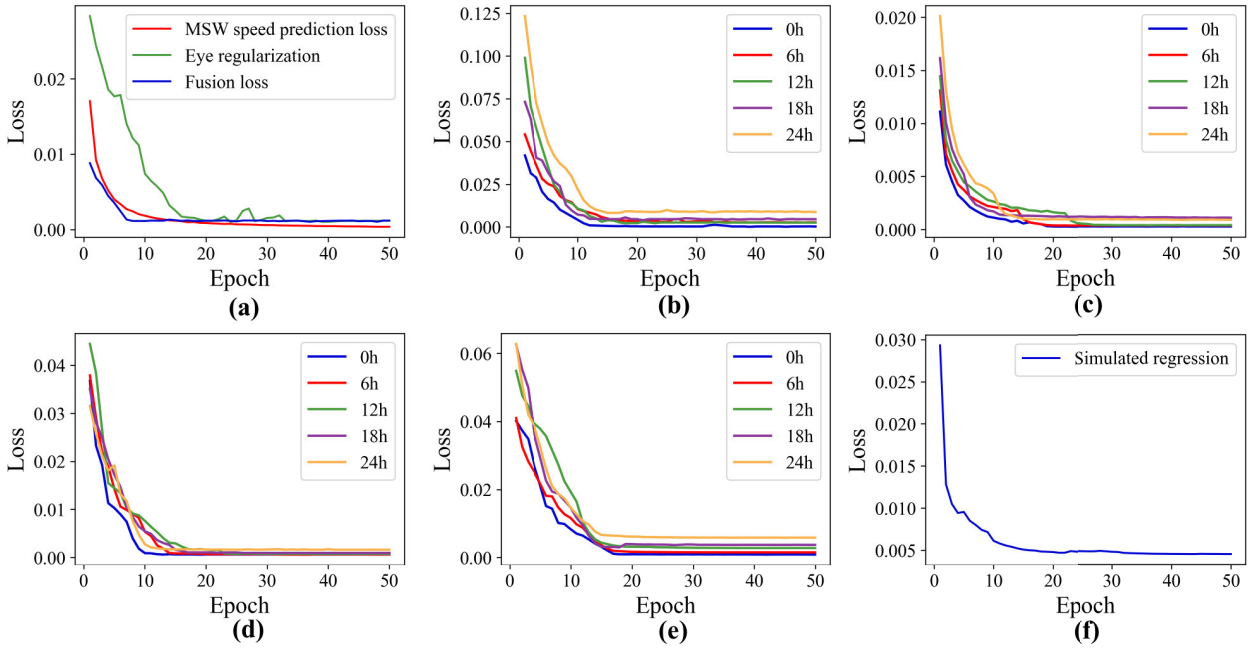


Fig. 16. Convergence curves of (a) MSW speed prediction loss, the eye regression loss, and the fusion loss of Pici-Net on FY4A-TC, (b) overall loss of Pici-Net on FY4A-TC, (c) overall loss of Pici-Net+ on FY4A-TC, (d) overall loss of Pici-Net+ on TCIR, (e) overall loss of Pici-Net++ on HURSAT-B1, and (f) simulation loss of PSM in Pici-Net++ on HURSAT-B1. The horizontal axis represents the training epochs and the vertical axis represents the loss values.

experts in making decisions by multisource information to reduce prediction errors. For Pici-Net++, we analyze the correlations between the simulated parameters and the reference MSW speed. As can be seen in Fig. 10(d), the simulated sea level pressure and the simulated TC translation speed show strong correlations with the reference MSW speed. It suggests that the parameters simulated by PSM are valid and can be used for MSW speed prediction. Besides, the textural features learned by Pici-Net or Pici-Net+ are only weakly correlated to the sea level pressure or the TC translation speed, as both parameters used in these models are derived from track data. Meanwhile, Pici-Net++ obtains the physical parameters through the smart simulation by PSM which leverages the extracted textural features instead of the track data. Thus, the simulated parameters are strongly correlated to the textural features, validating the interpretability of PSM.

4) Ablation: There are two parts to the ablation study, the architecture ablation and the parameter ablation. The former is to demonstrate the efficacy of each major module of the proposed Pici-Nets while the latter is to justify the value of the physical parameters in TC intensity prediction. The overall performance of each model is reflected by the MSW speed prediction errors and STDs as presented in Tables V–VII. The detailed analysis is given as follows.

a) Architecture ablation: As shown in Tables V–VII, every original Pici-Net outperforms all of its ablated models. The results verify the value of each module being removed from Pici-Nets. For example, the ablated model without the spiral extractor cannot directly learn variable spiral patterns, which, however, are strongly related to TC intensity. Thus, the fact that they are outperformed by Pici-Nets verifies the efficacy of the spiral extractor. The model without the eye extractor cannot analyze the textural information of the eye regions. Meanwhile, GLCM in Pici-Nets can leverage the spectral–spatial information in the MSIs to depict the textural details of the TC structures as illustrated in Fig. 7, thereby improving the physical explainability of the proposed model. The model without MPDF cannot leverage the multimodal information provided by MSIs and TC track data to make thorough judgments. Without processing the motional priors or the environmental factors, the ablated models experience a decay in prediction performance and loss in physical interpretability, compared to the proposed Pici-Nets.

Specifically, Table V shows that Pici-Net outperforms the ablated model "with CA," demonstrating the effectiveness of the self-attention mechanism [40] in Pici-Net. This attention block enables Pici-Net to explore global spectral–spatial features and recognize TC structures without being interfered with by local noise, thus providing essential information for TC intensity prediction and physical parameter simulation.

Table VII shows that Pici-Net++ outperforms its variation trained in "single-stage." It reveals the efficacy of the dual-stage training strategy adopted by Pici-Net++, which allows PSM to accurately simulate the physical parameters without being influenced by the intensity prediction errors. Accuracy and stability, which make up the reliability of the simulated data, are crucial to TC MSW speed prediction. Meanwhile, the STD of the "single-stage" model is significantly higher than that of the original Pici-Net++, indicating the former is less stable than the latter. This is because it is difficult for Pici-Net++ to keep the balance between the simulation net PSM and the prediction net Pici-Net++ when trained simultaneously in a single stage. The errors of these two modules affect each other, degrading the performance of both and thus producing unstable results.

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b) Physical data ablation: As can be seen in Tables V–VII, the models without the TC translation speed lack reference basis for the positional changes of TCs, leading to diminished ability to perceive MSW speed variations in the prediction period. The models without using the sea level pressure at the cyclone center lack the information of the cyclone center environment, inevitably lowering the prediction accuracy of MSW speed. The models without using any of these physical parameters denoted as "w/o S+P" cannot recognize the initial and the changing status of TCs, limiting their ability to track MSW speed. These results show the importance of PSM for TC forecast with incomplete data.

Notably, there is one ablated model of Pici-Net++ denoted by "with P_e" that uses the minimum sea level pressure $P_e(\underline{\mathbf{X}}_{i,t}^{(\text{Target})})$ estimated by the traditional statistical method [48] formulated in (28) rather than being simulated by PSM. To justify the reliability of the statistical model, we compare the estimated data $P_e(\underline{\mathbf{X}}_{i,t}^{(\text{Target})})$ with the reference data in Fig. 11. As can be seen, for both of the FY4A-TC and TCIR datasets, given any reference MSW speed, the minimum sea level pressure estimated by (28) with is very close to that provided by the best-track data, with an error below 15.5 hPa. Moreover, the trajectory curve reflecting how the estimated pressure varies with respect to the reference MSW speed and the curve of the dynamics of the reference data are very much alike in each plot. These results show that the model in (28) can reliably estimate sea level pressure. However, as shown by Table VII, Pici-Net++ outperforms the ablated model "with P_e," indicating that PSM provides more valuable simulated motional priors and environmental factors for the prediction task than the statistical estimation method, thus demonstrating the efficacy of PSM and the reliability of the simulated data. With such ability, the proposed Pici-Nets can be applied to various incomplete TC datasets utilizing the simulation model.

5) Generalizability on Different Basins: Meteorologists often study TCs in relation to the ocean basins where they occur, since TC evolvement is under the impact of the environment. Thus, it is interesting to see how the TC MSW prediction models perform in each basin Instead of designing one model for each basin, we apply Pici-Nets to all the six basins and analyze their generalizability. Specifically, we conduct two sets of experiments, 1) training and testing Pici-Nets on the TC data in a single basin and 2) training Pici-Nets on some of the basins and testing them on a different basin to show the performance across the basins. The train-test data split for TCIR and HURSAT-B1 is described in Section IV-A. The results are presented in Figs. 12–15. Each polygon in Figs. 12 and 13 visualizes the MSW-speed prediction performance given by one method over a prediction time. Fig. 12 shows the prediction ability of Pici-Net+ exhibited for each basin covered by the TCIR dataset. Fig. 13 demonstrates the cross-basin prediction ability of Pici-Net++ for the HURSAT-B1 dataset, The results indicate that the temporal structure representation and multimodal data fusion modules shared by Pici-Nets can learn variable characteristics of TC samples from multiple ocean basins and generalize to TC samples in other basins. Fig. 14 shows the majority of the absolute errors yielded by Pici-Nets falling within the range of 0–10 m/s in the single-basin evaluation, which means that Pici-Nets perform stably well under the guidance of the general physical knowledge of TCs, regardless of the characteristics of different basins. Fig. 15 also show that Pici-Nets predict with low errors across different basins, reflecting the generalizability of the proposed models.

6) Efficiency: Fig. 16(a) shows the curves of the eye regression loss in (7), the fusion loss in (9), the MSW speed prediction loss in (19) and the overall loss in (20) of Pici-Net on FY4A-TC. Fig. 16(b)–(e) shows the curves of the overall loss of Pici-Net+ and Pici-Net++ on variable datasets. Fig. 16(f) shows the simulation loss in (25) of PSM on the source data, i.e., FY4A-TC. As training time increases, all the loss curves exhibit a decreasing trend, ultimately leveling off and converging within 20 epochs. These results validate the feasibility and efficiency of the proposed models.

V. Discussion

Although Pici-Nets have achieved promising results in the MSW speed prediction, there are still rooms for improvement. From Tables II–VI, it can be observed that the prediction error increases with respect to the prediction time. This is because TCs are strong nonlinear systems, which can develop rapidly with the intensity changing abruptly. It might be helpful to augment the TC samples with abrupt changes during training to learn more from the temporal information. As shown by Figs. 14 and 15, prediction outliers occur in every test basin. This may be caused by the bias of sampling angles of different satellites and the variations of TC structures in each basin. A possible solution is to train the models with a small number of historical samples captured by different satellites from each basin and force the model to learn the similarity of these samples to adapt to cross-basin variations.

VI. Conclusion

In this work, we propose a family of novel physics-informed networks, namely, Pici-Nets, for TC MSW forecast. These models can mimic human experts in making decisions with various aspects of information of TCs. Guided by the structural knowledge of TCs, Pici-Nets can reconstruct the textural details of the eye regions and the cloud spirals of TCs that are closely related to MSW speed. Moreover, PSM is introduced to improve the interpretability and generalizability of Pici-Nets by generating the missing physical data for model training and MSW speed prediction. The results validate the effectiveness, explainability, interpretability, generalizability, and efficiency of the proposed methods, as Pici-Nets give SOTA performances on one self-collected TC MSI dataset and two sets of public datasets in 0–24-h TC intensity prediction.

References

[1] M. A.-A. Hoque, S. Phinn, C. Roelfsema, and I. Childs, "Tropical cyclone disaster management using remote sensing and spatial analysis: A review," *Int. J. Disaster Risk Reduction*, vol. 22, pp. 345–354, Jun. 2017.

[2] B. Chen, B. F. Chen, and Y. N. Chen, "Real-time tropical cyclone intensity estimation by handling temporally heterogeneous satellite data," in *Proc. AAAI Conf. Artif. Intell.*, 2021, vol. 35, no. 17, pp. 14721–14728.

[3] B.-F. Chen, B. Chen, H.-T. Lin, and R. L. Elsberry, "Estimating tropical cyclone intensity by satellite imagery utilizing convolutional neural networks," *Weather Forecasting*, vol. 34, no. 2, pp. 447–465, Apr. 2019.

Authorized licensed use limited to: Donghua University. Downloaded on March 05,2025 at 02:35:43 UTC from IEEE Xplore. Restrictions apply.

[4] M. Dawood, A. Asif, and F. U. A. A. Minhas, "Deep-PHURIE: Deep learning based hurricane intensity estimation from infrared satellite imagery," *Neural Comput. Appl.*, vol. 32, no. 13, pp. 9009–9017, Jul. 2020.

[5] C.-J. Zhang, X.-J. Wang, L.-M. Ma, and X.-Q. Lu, "Tropical cyclone intensity classification and estimation using infrared satellite images with deep learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 2070–2086, 2021.

[6] Z. Chen and X. Yu, "A novel tensor network for tropical cyclone intensity estimation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3226–3243, Apr. 2021.

[7] Y. Xiu, Z. Chen, X. Pu, H. Bi, and F. Xu, "Tropical cyclone intensity prediction by spectral–temporal dislocation and attention-based networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Pasadena, CA, USA, Jul. 2023, pp. 4931–4934.

[8] W. Mei, C. Pasquero, and F. Primeau, "The effect of translation speed upon the intensity of tropical cyclones over the tropical ocean," *Geophys. Res. Lett.*, vol. 39, no. 7, Apr. 2012. [Online]. Available: https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2011GL050765

[9] L. Xiao, P. F. Liu, and H. Li, "Progress and challenges in the fusion of multisource spatial–spectral remote sensing images," *J. Image Graph.*, vol. 25, no. 5, pp. 851–863, 2020.

[10] Q. Jin, X. Fan, J. Liu, Z. Xue, and H. Jian, "Using eXtreme gradient BOOSTing to predict changes in tropical cyclone intensity over the Western North Pacific," *Atmosphere*, vol. 10, no. 6, p. 341, Jun. 2019.

[11] R. L. Bankert and P. M. Tag, "An automated method to estimate tropical cyclone intensity using SSM/I imagery," *J. Appl. Meteorol.*, vol. 41, no. 5, pp. 461–472, May 2002.

[12] L. Boussioux, C. Zeng, T. Guénais, and D. Bertsimas, "Hurricane forecasting: A novel multimodal machine learning framework," *Weather Forecasting*, vol. 37, no. 6, pp. 817–831, Jun. 2022.

[13] W. Li, Y. Li, Y. Yang, D. Song, Z. Wei, and A.-A. Liu, "Multimodal adversarial fusion for typhoon intensity forecasting," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4102612.

[14] J. C. Sadler, "Tropical cyclones of the eastern North Pacific as revealed by TIROS observations," *J. Appl. Meteorol.*, vol. 3, no. 4, pp. 347–366, Aug. 1964.

[15] B. Chen, B.-F. Chen, and H.-T. Lin, "Rotation-blended CNNs on a new open dataset for tropical cyclone image-to-intensity regression," in *Proc. 24th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Jul. 2018, pp. 90–99.

[16] J. P. Kossin, "New global tropical cyclone data set from ISCCP B1 geostationary satellite observations," *J. Appl. Remote Sens.*, vol. 1, no. 1, Feb. 2007, Art. no. 013505.

[17] Y. Dong et al., "FY-4 Meteorological Satellite," *Aerosp. China*, vol. 18, no. 2, pp. 31–39, Feb. 2017.

[18] V. F. Dvorak, "A technique for the analysis and forecasting of tropical cyclone intensities from satellite pictures," Nat. Ocean. Atmos. Admin., Washington, DC, USA, Tech. Rep. Revision of NOAA TM NESS 36, 1973. [Online]. Available: https://repository.library.noaa.gov/view/noaa/18546

[19] V. F. Dvorak, "Tropical cyclone intensity analysis and forecasting from satellite imagery," *Monthly Weather Rev.*, vol. 103, no. 5, pp. 420–430, May 1975.

[20] V. F. Dvorak, "Tropical cyclone intensity analysis and forecasting from satellite visible or enhanced infraredimagery," *NESS*, vol. 103, no. 5, pp. 420–430, 1982.

[21] V. F. Dvorak, "Tropical cyclone intensity analysis using satellite data," U.S. Dept. Commerce, Nat. Ocean. Atmospheric Admin., Nat. Environ. Satell., Data, Inf. Service, Washington, DC, USA, Tech. Rep. NOAA–S/T 84-232, 1984. [Online]. Available: https://repository.library.noaa.gov/view/noaa/19322

[22] V. F. Dvorak, "A workbook on tropical clouds and cloud systems observed in satellite imagery," US Dept. Commerce, Nat. Ocean. Atmos. Admin., Nat. Environ. Satell., Data, Inf. Service, Nat. Weather Service, vol. 1, 1990.

[23] C. S. Velden, T. L. Olander, and R. M. Zehr, "Development of an objective scheme to estimate tropical cyclone intensity from digital geostationary satellite infrared imagery," *Weather Forecasting*, vol. 13, no. 1, pp. 172–186, Mar. 1998.

[24] T. L. Olander, C. S. Velden, and M. A. Turk, "Development of the advanced objective Dvorak technique (AODT)-current progress and future directions," in *Proc. 25th Conf. Hurricanes Tropical Meteorology*, San Diego, CA, USA, 2002, pp. 585–586.

[25] T. L. Olander and C. S. Velden, "The advanced Dvorak technique: Continued development of an objective scheme to estimate tropical cyclone intensity using geostationary infrared satellite imagery," *Weather Forecasting*, vol. 22, no. 2, pp. 287–298, Apr. 2007.

[26] C. P. Guard, L. E. Carr, F. H. Wells, R. A. Jeffries, N. D. Gural, and D. K. Edson, "Joint typhoon warning center and the challenges of multibasin tropical cyclone forecasting," *Weather Forecasting*, vol. 7, no. 2, pp. 328–352, Jun. 1992.

[27] X. Zhang et al., "Representing multiple scales in the Hurricane weather research and forecasting modeling system: Design of multiple sets of movable multilevel nesting and the basin-scale HWRF forecast application," *Weather Forecasting*, vol. 31, no. 6, pp. 2019–2034, 2016.

[28] P. Bauer, P. Lopez, E. Moreau, F. Chevallier, A. Benedetti, and M. Bonazzola, "The European centre for medium-range weather forecasts global rainfall data assimilation experimentation," in *Measuring Precipitation From Space*. Dordrecht, The Netherlands: Spinger, 2007, pp. 447–457. [Online]. Available: https://link.springer.com/book/10.1007/978-1-4020-5835-6

[29] E. N. Rajagopal, M. D. Gupta, V. S. Prasad, and S. Mohandas, "Overview of global data assimilation-forecast system," in *Proc. ORT*, 2011, p. 1.

[30] B. R. Jarvinen and C. J. Neumann, *Statistical Forecasts of Tropical Cyclone Intensity for the North Atlantic Basin*. Washington, DC, USA: National Oceanic and Atmospheric Administration, 1979. [Online]. Available: https://repository.library.noaa.gov/view/noaa/6555

[31] J. A. Knaff, M. DeMaria, C. R. Sampson, and J. M. Gross, "Statistical, 5-Day tropical cyclone intensity forecasts derived from climatology and persistence," *Weather Forecasting*, vol. 18, no. 1, pp. 80–92, Feb. 2003.

[32] M. DeMaria and J. Kaplan, "A statistical hurricane intensity prediction scheme (SHIPS) for the Atlantic basin," *Weather Forecasting*, vol. 9, no. 2, pp. 209–220, Jun. 1994.

[33] M. DeMaria and J. Kaplan, "An updated statistical hurricane intensity prediction scheme (SHIPS) for the Atlantic and Eastern North Pacific basins," *Weather Forecasting*, vol. 14, no. 3, pp. 326–337, Jun. 1999.

[34] J. M. Fritsch and C. F. Chappell, "Numerical prediction of convectively driven mesoscale pressure systems. Part I: Convective parameterization," *J. Atmos. Sci.*, vol. 37, no. 8, pp. 1722–1733, Aug. 1980.

[35] J. M. Fritsch and C. F. Chappell, "Numerical prediction of convectively driven mesoscale pressure systems. Part II. Mesoscale model," *J. Atmos. Sci.*, vol. 37, no. 8, pp. 1734–1762, Aug. 1980.

[36] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 25, 2012, pp. 1–9.

[37] C. Wang, Y. Xiu, and Z. Chen, "Cyclone intensity prediction via multispectral image sharpening," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[38] C. Huang, C. Bai, S. Chan, and J. Zhang, "MMSTN: A multi-modal spatial–temporal network for tropical cyclone short-term prediction," *Geophys. Res. Lett.*, vol. 49, no. 4, Feb. 2022, Art. no. e2021GL096898.

[39] P. Wang, P. Wang, D. Wang, and B. Xue, "A conformal regressor with random forests for tropical cyclone intensity estimation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4105914.

[40] J. Fu et al., "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3141–3149.

[41] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, Munich, Germany. Dordrecht, The Netherlands: Springer, Oct. 2015, pp. 234–241.

[42] S. Mu, X. Li, and H. Wang, "The fusion of physical, textural, and morphological information in SAR imagery for hurricane wind speed retrieval based on deep learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4207513.

[43] K. V. Sasidhar, T. K. Chaitanya, and A. P. Kesarkar, "Application of image retrieval techniques to understand evolving weather," *Int. J. Sci. Res. Develop.*, vol. 2, no. 7, pp. 570–573, 2014.

[44] A. K. Shakya, A. Ramola, Y. Tandon, and A. Vidhyarthi, "Suomi NPP satellite data processing to estimate impact of cyclone fani on LU/LC," in *Proc. 10th Int. Conf. Comput., Commun. Netw. Technol. (ICCCNT)*, Jul. 2019, pp. 1–5.

[45] X. Zhang, W. Huang, Q. Wang, and X. Li, "SSR-NET: Spatial–spectral reconstruction network for hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5953–5965, Jul. 2021.

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[46] G. D. Atkinson and C. R. Holliday, "Tropical cyclone minimum sea level pressure/maximum sustained wind relationship for the Western North Pacific," *Monthly Weather Rev.*, vol. 105, no. 4, pp. 421–427, Apr. 1977.
[47] D. E. Rumelhart, G. E. Hinton, and R. J. Williams, "Learning representations by back-propagating errors," *Nature*, vol. 323, no. 6088, pp. 533–536, Oct. 1986.
[48] M. Ying et al., "An overview of the China meteorological administration tropical cyclone database," *J. Atmos. Ocean. Technol.*, vol. 31, no. 2, pp. 287–301, Feb. 2014.
[49] X. Lu et al., "Western North Pacific tropical cyclone database created by the China meteorological administration," *Adv. Atmos. Sci.*, vol. 38, no. 4, pp. 690–699, Apr. 2021.
[50] K. R. Knapp, M. C. Kruk, D. H. Levinson, H. J. Diamond, and C. J. Neumann, "The international best track archive for climate stewardship (IBTrACS): Unifying tropical cyclone data," *Bull. Amer. Meteorological Soc.*, vol. 91, no. 3, pp. 363–376, Mar. 2010, doi: 10.1175/2009bams2755.1.
[51] J. Gahtan, K. R. Knapp, C. J. Schreck, H. J. Diamond, J. P. Kossin, and M. C. Kruk, "International best track archive for climate stewardship (IBTrACS) project, version 4r01. [indicate subset used]," NOAA Nat. Centers Environ. Inf., Tech. Rep., 2024, doi: 10.25921/82ty-9e16. [Online]. Available: https://www.ncei.noaa.gov/products/international-best-track-archive
[52] X. Wang, Y. Feng, R. Song, Z. Mu, and C. Song, "Multi-attentive hierarchical dense fusion net for fusion classification of hyperspectral and LiDAR data," *Inf. Fusion*, vol. 82, pp. 1–18, Jun. 2022.
[53] M. Franzese and A. Iuliano, "Correlation analysis," in *Encyclopedia of Bioinformatics and Computational Biology: ABC of Bioinformatics*, vol. 1. Amsterdam, The Netherlands: Elsevier, 2018, pp. 706–721.



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